

Geometry- and topology-controlled synchronization phase transition on manifolds*

Yang Tian^{1,†}

¹*Infplane Computing Technologies Ltd, Beijing, 100080, China*

In this work, we explore how the geometry and topology of the underlying manifold shape the synchronization phase transition of a system. To do so, we extend the Kuramoto-Sakaguchi model from spheres to compact, connected, orientable, and homogeneous Riemannian manifolds of arbitrary dimension. Starting from the mean-field kinetic equation on the manifold, we derive a local response equation for the order parameter near the incoherent state and separate the geometric and topological contributions to the phase transition out of the incoherent state. The manifold geometry determines the averaged projection factor $\kappa(M)$, which directly controls the coupling strength required to destabilize the incoherent state. The critical coupling is determined jointly by this geometric factor and the response of the intrinsic drift fields. The manifold topology affects the phase transition through the Euler characteristic $\chi(M)$: the Poincaré-Hopf relation fixes the net defect charge of the incipient ordered texture, and the local reduction and sign conditions stated below allow the same Euler-characteristic data to constrain the cubic coefficient of the reduced response equation. In that conditional class, a non-zero Euler characteristic gives the cubic sign used in the reduced normal form, and an additional stabilization condition gives a discontinuous local transition. When $\chi(M) = 0$, the local branch is determined by the normal-form coefficients rather than by the Euler characteristic alone. We evaluate these geometric and topological indicators on representative families including hyperspheres, sphere products, complex Grassmannians, complex projective spaces, flat tori, real Stiefel manifolds, rotation groups, and unitary groups. Our framework recovers the topological part of the classical hyperspherical parity distinction and extends it to a broad class of non-spherical state spaces. It therefore identifies the geometric factor that controls the coupling strength required for the loss of stability of the incoherent state, the unconditional topological defect constraint, and the conditional topological selection of the local phase transition scenario in diverse synchronization phenomena.

I. INTRODUCTION

The Kuramoto model serves as a prototypical framework for investigating low-dimensional synchronization phenomena [1–4]. Its generalized form, the Kuramoto-Sakaguchi model, also plays a crucial role in studying both synchronization and asynchronization in low-dimensional systems [2, 3, 5]. However, to accurately describe a broader range of biological, chemical, and physical systems [6–8], it is necessary to consider high-dimensional settings ($D \geq 3$), in which the original low-dimensional formulations ($D \leq 2$) of the Kuramoto model and the Kuramoto-Sakaguchi model become inadequate [9–11]. Moreover, the phase space of a high-dimensional system need not be a standard D -dimensional sphere. In that case, high-dimensional extensions built specifically on the hypersphere may fail to represent the underlying state space [12–16].

In this study, we extend the Kuramoto-Sakaguchi model to compact, connected, orientable, and homogeneous D -dimensional Riemannian manifolds, which may possess non-trivial topology or an arbitrary number of holes [17]. This class already contains the principal geometric examples relevant to synchronization, including spheres, flat tori, Stiefel manifolds, rotation groups, and unitary groups [9, 12–14, 18–22]. Related Lohe-type and

unitary-group models provide a complementary matrix-valued route to manifold synchronization [14, 23–33]. The original Kuramoto model, the original Kuramoto-Sakaguchi model, and the generalized Kuramoto model on the hypersphere all serve as special cases of our extended model.

Our analysis yields three main conclusions and Table I summarizes how the theoretical framework is built. First, the manifold geometry determines the averaged tangent-projection factor $\kappa(M)$. This factor multiplies the coupling-induced source term in the linearized kinetic equation. The critical coupling K_c is then determined jointly by this geometric factor and by the response of the intrinsic drift fields. In scalar reductions where the transport response factorizes as $s_M(\lambda; h) = \kappa(M) \Xi(\lambda; h)$, the scalar factor $\Xi(\lambda; h)$ is the intrinsic-drift response and the critical coupling is $K_c = 1/[\kappa(M) \Xi(0^+; h)]$ for $\alpha = 0$. Second, the manifold topology gives an unconditional defect constraint and a conditional constraint on local phase transition scenarios. The unconditional statement is the Poincaré-Hopf relation: the total index of the zeros of the critical tangent field equals the Euler characteristic [34], so a non-zero Euler characteristic imposes a non-zero net defect charge on the critical texture. The phase transition statement additionally uses the locality conditions of Appendix D and the sign condition stated in Appendix E. Under those assumptions, a non-zero Euler characteristic gives the cubic sign used in the one-mode normal form; if the stabilizing condition $\Lambda_5 > 0$ also holds, the local transition is discontinuous. When $\chi(M) = 0$, the local branch is determined by the normal-form co-

* Correspondence should be addressed to Yang Tian.

† tyanyang04@gmail.com & yang.tian@infplane.com

Section and step	What we derive	How the result is used
In Sec. II, manifold Kuramoto-Sakaguchi model	We place the oscillator state on a compact homogeneous manifold M , fix an embedding $F : M \hookrightarrow \mathbb{R}^{D_a}$, and project the coupling term onto the tangent space.	This gives a common dynamical model for spherical and non-spherical state spaces and specifies where geometry can enter the coupling.
In Sec. II D, mean-field kinetic equation	We derive the density equation on the tangent bundle and expand it near the incoherent state.	The expansion separates the response into a linear spectral problem and a nonlinear local response equation.
In Sec. III B, linear response and geometry	We linearize the kinetic equation and evaluate the coupling-induced source term in the chosen order parameter mode.	The averaged tangent-projection factor $\kappa(M)$ gives the geometric contribution to the critical coupling K_c , together with the response of the intrinsic drift fields.
In Secs. III C–E, nonlinear response and topology	We reduce the local order parameter response to a one-mode normal form and use the Poincaré-Hopf relation for the critical tangent field.	The Euler characteristic gives an unconditional defect-charge constraint. Under the locality and sign conditions, it also gives the cubic sign used in the reduced local response.
In Sec. IV, representative manifolds and numerical tests	We compute $\kappa(M)$, $\chi(M)$, defect constraints, and local transition indicators on representative spherical and non-spherical manifolds.	These calculations and simulations test how geometry determines K_c and how topology constrains defects and, conditionally, the local transition scenario.

TABLE I. Logical structure of the present work. The table follows the route used in the paper: we first formulate the Kuramoto-Sakaguchi model on a general state manifold, then derive the kinetic equation, separate the linear and nonlinear response terms near the incoherent state, and finally identify the geometric and topological information that enters the synchronization phase transition.

efficients rather than by the Euler characteristic alone. Here the sign condition means that the contribution of neighborhoods of the zeros weighted by their local indices has a definite sign and dominates the remaining part of the cubic density that is not weighted by the indices. Third, these geometric, spectral, and topological ingredients control physical properties that are directly relevant to statistical physics. They determine the critical coupling for the loss of stability of the incoherent state, the conditional phase transition type, the possible presence of hysteresis, and the defect content of the incipient ordered texture, and we compute these indicators on representative manifolds, including hyperspheres, sphere products, complex Grassmannians, complex projective spaces, flat tori, real Stiefel manifolds, rotation groups, and unitary groups [22, 35].

The present framework also provides a basis for several further statistical-physics problems on high-dimensional state spaces. It can be used to study finite-size critical fluctuations near the loss of stability of the incoherent state, defect kinetics and charge balance in the emerging ordered texture, metastability and noise-induced switching on the discontinuous side, and nonequilibrium hysteresis under slow or fast parameter ramps. It also opens a route to systematic comparisons among multi-angle oscillators, matrix synchronization models, and other manifold-valued oscillator systems within a common geometric and topological language. In this sense, the theory identifies which part of the loss of stability of the incoherent state is determined by geometry, which part of the

nonlinear response is constrained by topology, and which physical properties remain to be selected by the analytic structure of the underlying model.

II. GENERAL MODEL ON THE MANIFOLD

A. Geometric setting

In this work, we consider a smooth, connected, compact, orientable, and homogeneous Riemannian manifold (M, g) of dimension D without boundary. The state of every oscillator i is defined on this manifold, $\sigma_i \in M$. The Riemannian metric g induces the Levi-Civita connection ∇_M and the volume measure μ [17, 36]. For any $\sigma_i \in M$, we denote by $T_{\sigma_i}M$ the tangent space at σ_i , equipped with the inner product $\langle \cdot, \cdot \rangle_g$ and the norm $\| \cdot \|_g$ [36].

According to Nash's embedding theorem [37, 38], there exists an isometric embedding $F_0 : (M, g) \hookrightarrow (\mathbb{R}^{D_0}, \langle \cdot, \cdot \rangle)$. In this work, we define a smooth embedding

$$F : M \hookrightarrow \mathbb{R}^{D_a}, \quad D_a \geq D_0 \quad (1)$$

and identify M with its image $F(M) \subset \mathbb{R}^{D_a}$. The differential $dF_{\sigma_i} : T_{\sigma_i}M \rightarrow \mathbb{R}^{D_a}$ is injective for each $\sigma_i \in M$. In some special cases (e.g., the hypersphere $M = S^D$), we additionally set $|F(\sigma_i)| = 1$ for all $\sigma_i \in M$ such that $F(M)$ lies on the unit sphere.

Through the embedding, we realize each tangent space

$T_{\sigma_i}M$ as a linear subspace of $\mathbb{R}^{D_a}$. We denote

$$\mathbf{P}_{\sigma_i}^\perp : \mathbb{R}^{D_a} \rightarrow T_{\sigma_i}M \quad (2)$$

as the orthogonal projection with respect to the Euclidean inner product [36]. This projection ensures that all vector fields defining the dynamics remain tangent to the manifold M . The transport formulation introduced below still makes sense on a general compact orientable manifold, while the homogeneous assumption is used when we reduce the averaged geometric operator to a scalar factor in Sec. III B.

B. General model in discrete form

Based on the definitions presented above, we first introduce our general model in its discrete form. The Kuramoto-Sakaguchi model on the manifold is written as

$$\frac{d}{dt}\sigma_i = V_i(\sigma_i) + \frac{K}{N} \sum_{j=1}^N \Gamma(\sigma_i, \sigma_j; \alpha), \quad (3)$$

where K denotes the coupling strength, N is the number of oscillators, and α denotes the phase-lag effect. Each V_i is a smooth tangent vector field of oscillator i that satisfies

$$\nabla_M \cdot V_i = 0, \quad (4)$$

which is the natural condition ensuring that the spatially uniform incoherent state used below remains stationary under the intrinsic drift. We define the function Γ as

$$\Gamma(\sigma_i, \sigma_j; \alpha) = \mathbf{P}_{\sigma_i}^\perp [(\mathbf{I} - \alpha A) F(\sigma_j)], \quad (5)$$

in which $\mathbf{P}_{\sigma_i}^\perp$ maps any vector in the ambient space $\mathbb{R}^{D_a}$ to the tangent space of M with respect to σ_i . The function F isometrically embeds oscillator states on M into the ambient space $\mathbb{R}^{D_a}$ to support the synchronization analysis in an Euclidean space. The anti-symmetric operator A is used to describe the non-reciprocal coupling caused by phase lags, whose strength is measured by α . We interpret $\mathbf{I} - \alpha A$ as a linearized phase-lag operator. If an exact finite phase lag is represented by the orthogonal operator $R_\alpha = \exp(-\alpha A)$, then $\mathbf{I} - \alpha A$ is its first-order truncation in α . We use this truncation to keep the coupling linear in the embedded order parameter. The operator $\mathbf{I} - \alpha A$ itself is not required to map $F(M)$ into $F(M)$ or to generate an exact motion on M ; the projection $\mathbf{P}_{\sigma_i}^\perp$ enforces tangency of the velocity at the receiving state σ_i . Thus, for every i , the right-hand side of Eq. (3) lies in $T_{\sigma_i}M$. The phase-lag term breaks the reciprocal structure when $|\alpha| > 0$, in analogy with the classic Kuramoto-Sakaguchi coupling on the circle [5].

In Eq. (5), $\mathbf{P}_{\sigma_i}^\perp F(\sigma_j)$ enforces synchronization among oscillators while $-\alpha \mathbf{P}_{\sigma_i}^\perp A F(\sigma_j)$ imposes asynchronization. The action of $\mathbf{P}_{\sigma_i}^\perp$ ensures that the change in oscillator state σ_i remains confined to the tangent space,

irrespective of whether the evolution of σ_i is governed by synchronizing or desynchronizing interactions. To analyze the global state of the system, the order parameter of the system is defined as

$$r(t) = \frac{1}{N} \sum_{i=1}^N F(\sigma_i), \quad (6)$$

which is the empirical mean of the embedded states $F(\sigma_i)$ in the ambient Euclidean space [1–3, 5]. The particular choice of embedding F determines which geometric features of the manifold are probed by the order parameter. In the classical examples, F is an isometric embedding into the unit sphere, so that $|r(t)| \in [0, 1]$ quantifies the degree of synchronization. As $|r|$ approaches 0, the system is in an asynchronization phase. As $|r|$ approaches 1, the system is in a synchronization phase.

C. Connection to existing models

For better conceptual clarity, we now explain the connection between the abstract model presented above and existing concrete models.

In a special case where (M, g) reduces to a standard sphere $S^{D-1} \subset \mathbb{R}^D$, we actually require $D_a = D$, $F(\sigma_i) = \sigma_i$, and $\|\sigma_i\| = 1$ (i.e., on the unit sphere). Meanwhile, the projection operator reduces to $\mathbf{P}_{\sigma_i}^\perp = \mathbf{I} - \sigma_i \sigma_i^\top$ due to changes on the Riemannian metric. Moreover, the natural frequency field V_i reduces to the generator of rotation Ω_i such that $V_i(\sigma_i) \rightarrow \Omega_i \sigma_i$. These changes enable us to rewrite Eqs. (3, 5) as

$$\begin{aligned} \frac{d}{dt}\sigma_i &= \Omega_i \sigma_i + \\ &\frac{K}{N} \sum_{j=1}^N [\sigma_j - (\sigma_i \cdot \sigma_j) \sigma_i - \alpha (A \sigma_j - (\sigma_i \cdot A \sigma_j) \sigma_i)], \end{aligned} \quad (7)$$

which is the extended Kuramoto-Sakaguchi model on the D -dimensional hypersphere. If we further set $\alpha = 0$, there is no phase-lag effect and we obtain the extended Kuramoto model on the D -dimensional hypersphere

$$\frac{d}{dt}\sigma_i = \Omega_i \sigma_i + \frac{K}{N} \sum_{j=1}^N [\sigma_j - (\sigma_i \cdot \sigma_j) \sigma_i], \quad (8)$$

which contains only synchronization forces. The gradient flow structure under this condition is elaborated in Appendix A. When $D = 2$, Eqs. (7-8) reduce to the original Kuramoto-Sakaguchi model and the original Kuramoto model, respectively.

In more general cases, the manifold formulation can recover different Kuramoto-type models as special cases. Different choices of the manifold M and the embedding F correspond to different state spaces and interaction geometries, while the different value of the phase-lag parameter α introduces different Kuramoto-Sakaguchi type

Model	M	F	Phase-lag
Kuramoto on S^1	S^1	$(\cos \sigma, \sin \sigma)$	$A = 0, \alpha = 0$
Kuramoto-Sakaguchi on S^1	S^1	$(\cos \sigma, \sin \sigma)$	$A = J, \text{ small } \alpha $
Kuramoto on S^{D-1}	S^{D-1}	Inclusion $F(\sigma) = \sigma$	$A = 0, \alpha = 0$
Frustrated Kuramoto on S^{D-1}	S^{D-1}	$F(\sigma_i) = \sigma_i$	$A \in \mathfrak{so}(D)$
Vector-phase Kuramoto on $\mathbb{T}^d$	$\mathbb{T}^d$	Product of S^1 embeddings	Block-diagonal A
Kuramoto on Stiefel/rotation manifolds	$\text{St}(p, n), \text{SO}(n)$	Canonical matrix embedding	A acts by BX or XB
Lohe/unitary Kuramoto	$\text{U}(d), \text{SU}(d)$	$F(U) = U$ in $\mathbb{C}^{d \times d}$	A in $\mathfrak{u}(d)$
Network-distributed Kuramoto	Any of the above M	Corresponding F	$K \rightarrow K_{ij}$, same A

TABLE II. Representative special cases of the general Kuramoto-Sakaguchi model Eqs. (3, 5). In all the cases, the state of oscillator i is denoted by $\sigma_i \in M$, where M is a smooth compact Riemannian manifold. $F : M \hookrightarrow \mathbb{R}^{D_a}$ is a fixed smooth embedding into an ambient Euclidean space $\mathbb{R}^{D_a}$. The coupling term is $\Gamma(\sigma_i, \sigma_j; \alpha) = \mathbf{P}_{\sigma_i}^\perp [(\mathbf{I} - \alpha A) F(\sigma_j)]$, where $\mathbf{P}_{\sigma_i}^\perp$ denotes the orthogonal projection from $\mathbb{R}^{D_a}$ onto the tangent space $T_{\sigma_i} M$, $A \in \mathfrak{so}(D_a)$ is a fixed skew-symmetric matrix representing a linearized generator of rotations in the embedding space, and $\alpha \in \mathbb{R}$ is a scalar phase-lag parameter. In the table, each row corresponds to a particular choice of M , F and A (e.g., $M = S^1$ with $F(\sigma) = (\cos \sigma, \sin \sigma)$ and $A = 0$ in the first row, or $M = S^{D-1}$ with $F(\sigma_i) = \sigma_i$ and $A \in \mathfrak{so}(D_a)$ in the fourth row), which reproduces a special Kuramoto-type model proposed in previous studies [2, 3, 9–13, 15, 16, 32, 33, 39–41]. Please see Appendix B for the detailed definitions of all these special cases. Their associated mathematical notions are not elaborated in the main text since they are irrelevant to our subsequent analysis.

frustration. A few representative examples are summarized in Table II. Their definitions and relations with Eqs. (3, 5) are presented in Appendix B.

D. General model in continuous form

Building on the established correspondence with previous models, we proceed to develop the continuous counterpart of our discrete formulation. The continuum formulation arises in the mean-field limit $N \rightarrow \infty$ of Eqs. (3, 5), whose detailed derivation is presented in Appendix C [14, 42]. In general, we consider an empirical measure on M and relate it with the limit $N \rightarrow \infty$ in the weak formulation of the dynamics. We separate the distribution of intrinsic drift fields from the conditional spatial density and write the limiting measure on the tangent bundle as $\rho(\sigma, V, t) h(V) d\mu(\sigma) dV$, where $\rho(\sigma, V, t)$ denotes the conditional density of oscillators at time t , in state $\sigma \in M$, for a fixed intrinsic field V , and $h(V)$ is the time-independent probability density of V . The conditional density satisfies

$$\int_M \rho(\sigma, V, t) d\mu(\sigma) = 1 \quad (9)$$

for every fixed V , where TM is the tangent bundle of the manifold (i.e., the union of all the tangent spaces), μ denotes a volume measure, and $d\mu(\sigma) = \sqrt{\det g} d^D x$ is the Riemannian volume element [36, 38].

In the continuous setting, the order parameter of the system can be expressed as

$$r(t) = \int_{TM} \int_M F(\sigma) \rho(\sigma, V, t) h(V) d\mu(\sigma) dV, \quad (10)$$

which is the expectation of $F(\sigma)$ with respect to the probability measure $\rho(\sigma, V, t) d\mu(\sigma) h(V) dV$ (here $h(V)$ is the probability density of V at each position). To reflect the influence of the global coupling force, represented by the order parameter, on the evolution of oscillator states, we propose the velocity field equation of the oscillator states

$$\mathbf{v}(\sigma, V, t) = V(\sigma) + K \mathbf{P}_\sigma^\perp(r(t) - \alpha A r(t)). \quad (11)$$

In this equation, $V(\sigma)$ describes the intrinsic velocity that depends on the local position σ and reflects heterogeneity. For simplicity, we assume $\mathbb{E}[V(\sigma)] = 0$ and $\mathbb{E}[V(\sigma) \otimes V(\sigma)] = \Delta^2 \mathbf{I}$, where Δ measures the variance of intrinsic velocity and $\mathbf{I}$ is the identity tensor on the tangent space. $K \mathbf{P}_\sigma^\perp r(t)$ drives the system toward synchronization given the feedback effects of the order parameter. The non-reciprocal coupling term $-K \mathbf{P}_\sigma^\perp \alpha A r(t)$ models the reaction delays inherent in real systems and disrupts synchronization. The conservation law below uses only the fact that $\mathbf{v}(\sigma, V, t)$ is tangent to M , which is guaranteed by the projection $\mathbf{P}_\sigma^\perp$. It does not require $\mathbf{I} - \alpha A$ to be an exact rotation on the embedded manifold or to be divergence-free. The divergence-free assumption

in Eq. (4) is imposed on the intrinsic drift field V_i so that the incoherent uniform density remains stationary under the intrinsic transport.

The structure of the velocity field in Eq. (11) is consistent with the mean-field velocity obtained in the continuum limit. In Appendix C we show that, at the level of the limiting density, the drift term entering the continuity equation is given by an intrinsic part $V(\sigma)$ plus an interaction part obtained by averaging the discrete coupling $\Gamma(\sigma_i, \sigma_j; \alpha)$ over the distribution of oscillator states. When the interaction is expressed in terms of the order parameter $r(t)$ as in Eq. (10), this averaged coupling reduces exactly to the projection $P_\sigma^\perp(r(t) - \alpha A r(t))$ in Eq. (11).

The velocity field defined above effectively transports the probability density over the manifold to create the flow of a “probability fluid”. This probability flow describes how the statistical distribution of the oscillator system evolves over time in the phase space. In general, the evolution equation of this flow is written as

$$\frac{\partial}{\partial t} \rho + \nabla_M \cdot \mathbf{J} = 0, \quad (12)$$

where ∇_M is the previously defined covariant divergence operator and $\mathbf{J} = \rho \mathbf{v}$ is a probability density flow tensor [36]. For any $\sigma \in M$, $\mathbf{J}(\sigma)$ indicates the direction of probability flow and $|\mathbf{J}(\sigma)|$ quantifies the flow rate. The divergence $\nabla_M \cdot \mathbf{J}$ measures the net outflow per unit time and unit volume. There is a net outflow of probability density from the local volume element when $\nabla_M \cdot \mathbf{J} > 0$ and there is a net inflow of probability density when $\nabla_M \cdot \mathbf{J} < 0$. More precisely, the above equation can be rewritten in an explicit form under a local coordinate chart $\{x^1, \dots, x^D\}$ by explicitly calculating the covariant divergence

$$\frac{\partial}{\partial t} \rho + \frac{1}{\sqrt{\det g}} \frac{\partial}{\partial x^k} \left(\sqrt{\det g} \rho v^k \right) = 0, \quad (13)$$

in which v^k is the contravariant component of the velocity field (i.e., $\mathbf{v} = v^k \partial_k$) and $\rho v^k = \mathbf{J}^k$ is the contravariant component of the probability flow density. In Eq. (13), $\sqrt{\det g} \rho v^k$ measures the invariant probability flux density through a coordinate surface (i.e., the amount of probability crossing a unit coordinate area per unit time). $\sqrt{\det g}$ encodes the influence of the manifold curvature on the probability flow since $\det g$ varies sharply in regions of high curvature. When $\sqrt{\det g}$ enlarges, probability flow deceleration and density dilution occur, making curvature equivalent to a repulsive potential. When $\sqrt{\det g}$ reduces, probability flow accelerates and probability density concentrates, making curvature equivalent to an attractive potential. The normalization coefficient $\frac{1}{\sqrt{\det g}}$ normalizes this flux variation to unit volume.

Eqs. (12-13) interpret the probability flow as an incompressible fluid over the manifold and describe its local conservation law for every fixed intrinsic field V . The

corresponding global conservation law is

$$\frac{\partial}{\partial t} \int_{TM} \int_M \rho(\sigma, V, t) h(V) d\mu(\sigma) dV = 0. \quad (14)$$

The corresponding boundary condition is either that the manifold M is compact, or that a zero-flux condition (i.e., the Neumann-type condition) is imposed at the boundary. Subject to these constraints, the total probability density is conserved over time, with only local flows occurring.

III. GEOMETRY- AND TOPOLOGY-GOVERNED PHASE TRANSITION

A. Analysis framework

After extending the Kuramoto-Sakaguchi model to compact, connected, orientable, and homogeneous D -dimensional Riemannian manifolds, we present the derivation of our main theorem in this section. We separate the result into two parts. First, for the chosen embedding, we use the averaged tangent projection to define the coefficient $\kappa(M)$. This coefficient multiplies the coupling-induced source term in the linearized kinetic equation. We obtain the actual critical coupling from the full density-level spectral problem after the response of the intrinsic drift fields is evaluated, so the threshold is determined jointly by geometry and by the distribution of intrinsic drift fields. Second, we use the Euler characteristic to obtain the unconditional Poincaré-Hopf defect constraint on the critical tangent field. We then use the additional locality and sign conditions of Appendices D and E to obtain conditional statements about the cubic coefficient and the local phase transition scenario. In that conditional class, a non-zero Euler characteristic gives the cubic sign used by the one-mode normal form, and the further stabilizing condition $\Lambda_5 > 0$ gives a discontinuous local transition. When $\chi(M) = 0$, the local branch is determined by the normal-form coefficients rather than by the Euler characteristic alone. Below, we use the continuous form of the generalized model to elaborate our derivation. We first formulate the linear kinetic response of the order parameter under perturbations and then establish its connection with the geometric and topological characteristics of the manifold.

To implement this analysis, we consider a situation in which the system is initialized at a ground state

$$\rho_0(\sigma, V) = \frac{1}{\text{Vol}(M)} \quad (15)$$

for every fixed V (i.e., the spatial distribution is uniform over the manifold conditional on each intrinsic drift field, here $\text{Vol}(M)$ measures the total volume of M). This uniform distribution lies far from the synchronized state, leading to a near-zero value of the order parameter $r_0 \approx$

0. We add small perturbations on ρ_0 to affect r_0

$$\rho = \rho_0 + \delta_\rho, \quad r = r_0 + \delta_r. \quad (16)$$

These small perturbations can be used to model the influence of interactions between oscillators on the probability flow described in Eq. (12). Note that ρ_0 is a constant at every point on the manifold for each fixed V .

Under small perturbations, we first determine the linear stability of the incoherent state at the level of the full density perturbation. Once we identify a simple critical eigenvalue, we reduce the local nonlinear dynamics of the order parameter to a response equation of the form

$$\frac{\partial}{\partial t} r = \mathcal{L}(r) + \mathcal{N}(r), \quad (17)$$

where $\mathcal{L}(r)$ is the linear operator inherited from the kinetic spectral problem and $\mathcal{N}(r)$ is the non-linear term. In the following analysis, we identify the geometric source coefficient multiplying the coupling-induced source term, define the corresponding spectral threshold, and then relate the nonlinear coefficient to the topology of the manifold under the local reduction assumptions.

B. Linearized kinetic response and spectral threshold

a. Density-level linearization.- We first determine the linear response at the level of the full density perturbation. From Eq. (11), the perturbation of the velocity field is

$$\delta_{\mathbf{v}} = K \mathbf{P}_\sigma^\perp (\mathbf{I} - \alpha A) \delta_r. \quad (18)$$

We insert $\rho = \rho_0 + \delta_\rho$ and $\mathbf{v} = V + \delta_{\mathbf{v}}$ into the continuity equation Eq. (12) and keep only linear terms. This gives

$$\frac{\partial}{\partial t} \delta_\rho = \mathcal{A}_V [\delta_\rho] - K \rho_0 \nabla_M \cdot [\mathbf{P}_\sigma^\perp (\mathbf{I} - \alpha A) \delta_r], \quad (19)$$

where

$$\mathcal{A}_V [\varphi] = -\nabla_M \cdot (\varphi V) \quad (20)$$

is the transport generator associated with the intrinsic drift field. The order parameter perturbation is the first embedded moment

$$\delta_r = \int_{TM} \int_M F(\sigma) \delta_\rho(\sigma, V, t) h(V) d\mu(\sigma) dV. \quad (21)$$

Eqs. (19, 21) form the correct linearized problem. They are not closed as an autonomous equation for δ_r , because the transport term $\mathcal{A}_V [\delta_\rho]$ depends on the full density perturbation, not only on its first embedded moment.

b. Geometric source term.- Although the moment equation is not closed, we can still isolate the geometric part of the coupling-induced source term in Eq. (19). For every tangent vector field $\mathbf{X}$ on the compact manifold,

$$\int_M F(\sigma) \nabla_M \cdot \mathbf{X} d\mu(\sigma) = - \int_M dF_\sigma [\mathbf{X}(\sigma)] d\mu(\sigma). \quad (22)$$

Therefore, with $\mathbf{Y} = (\mathbf{I} - \alpha A) \delta_r$, we obtain the direct moment of the coupling-induced source term as

$$\begin{aligned} & -K \rho_0 \int_{TM} \int_M F(\sigma) \nabla_M \cdot [\mathbf{P}_\sigma^\perp \mathbf{Y}] h(V) d\mu(\sigma) dV \\ &= \frac{K}{\text{Vol}(M)} \int_M \mathbf{P}_\sigma^\perp \mathbf{Y} d\mu(\sigma). \end{aligned} \quad (23)$$

Up to this point, we have not used homogeneity. If the homogeneous embedding subspace probed by the order parameter is isotropic, then the averaged tangent projection acts as a scalar multiple of the identity on that subspace:

$$\frac{1}{\text{Vol}(M)} \int_M \mathbf{P}_\sigma^\perp \mathbf{Y} d\mu(\sigma) = \kappa(M) \mathbf{Y}. \quad (24)$$

We therefore identify $\kappa(M)$ as the averaged tangent-projection factor multiplying the coupling-induced source term,

$$\frac{K}{\text{Vol}(M)} \int_M \mathbf{P}_\sigma^\perp \mathbf{Y} d\mu(\sigma) = K \kappa(M) \mathbf{Y}. \quad (25)$$

This identity specifies the coupling-induced source term rather than a closed growth equation for δ_r ; the intrinsic-drift resolvent is evaluated separately.

c. Transport resolvent and self-consistency.- We note that the operator $\mathcal{A}_V$ is skew-adjoint on $L^2(M, \mu)$ when $\nabla_M \cdot V = 0$:

$$\int_M f \mathcal{A}_V [g] d\mu = - \int_M g \mathcal{A}_V [f] d\mu. \quad (26)$$

However, this skew-adjointness does not by itself determine the real part of the spectrum after the rank-finite mean-field coupling is added. We must obtain the loss of stability from the coupled density-moment problem. We therefore use a normal-mode ansatz $\delta_\rho = e^{\lambda t} \eta(\sigma, V)$ and $\delta_r = e^{\lambda t} q$. For $\text{Re } \lambda > 0$, Eq. (19) gives

$$(\lambda - \mathcal{A}_V) \eta = -K \rho_0 \nabla_M \cdot [\mathbf{P}_\sigma^\perp (\mathbf{I} - \alpha A) q]. \quad (27)$$

We apply the resolvent of the transport generator and impose the moment condition. This gives

$$q = K \mathcal{S}(\lambda; h) (\mathbf{I} - \alpha A) q, \quad (28)$$

where the linear susceptibility operator is

$$\begin{aligned} \mathcal{S}(\lambda; h) z &= - \frac{1}{\text{Vol}(M)} \int_{TM} \int_M F(\sigma) (\lambda - \mathcal{A}_V)^{-1} \\ &\quad \times \nabla_M \cdot [\mathbf{P}_\sigma^\perp z] h(V) d\mu(\sigma) dV. \end{aligned} \quad (29)$$

Geometric role	Detailed effect on the loss of stability of the incoherent state
Direct role in the linearized kinetic equation	Geometry determines how much of a mean-field perturbation becomes a tangent force on M . In an isotropic response mode, $\kappa(M)$ sets the geometric strength of the destabilizing coupling feedback: $\bar{\mathbf{P}} = \text{Vol}(M)^{-1} \int_M \mathbf{P}_\sigma^\perp d\mu(\sigma)$, and $\bar{\mathbf{P}} = \kappa(M) \mathbf{I}$ in the scalar isotropic reduction. The geometric coupling factor is then $K\kappa(M)$.
Role in the critical coupling	Geometry directly affects the coupling strength required to destabilize the incoherent state, but it does not determine the critical coupling alone. The critical coupling is fixed jointly by $\kappa(M)$ and the h -dependent susceptibility $\mathcal{S}(\lambda; h)$ through $\det[\mathbf{I} - K\mathcal{S}(\lambda; h)(\mathbf{I} - \alpha A)] = 0$. If $s_M(0^+; h) = \kappa(M) \Xi(0^+; h)$, then the scalar intrinsic-drift response is $\Xi(0^+; h)$ and $K_c = 1/[\kappa(M) \Xi(0^+; h)]$.

TABLE III. Geometric contribution to the loss of stability of the incoherent state for a fixed embedding. The coefficient $\kappa(M)$ is the isotropic averaged tangent-projection factor. It measures how strongly a mean-field perturbation is converted into a tangent coupling force on the manifold. The critical coupling is then determined by this geometric factor together with the intrinsic-drift response.

We therefore determine the critical coupling from the spectral self-consistency condition

$$\det[\mathbf{I} - K\mathcal{S}(\lambda; h)(\mathbf{I} - \alpha A)] = 0, \quad \text{Re } \lambda = 0. \quad (30)$$

Equivalently, we may approach the neutral spectrum by analytic continuation from $\text{Re } \lambda > 0$. In this formulation, the averaged projection factor $\kappa(M)$ is contained in the geometric source term in Eq. (25), whereas the actual threshold also depends on the intrinsic-drift distribution through the resolvent in Eq. (29). We define the intrinsic-drift response as the h -dependent susceptibility $\mathcal{S}(\lambda; h)$ in Eq. (29); when the scalar factorization $s_M(\lambda; h) = \kappa(M) \Xi(\lambda; h)$ holds, the corresponding scalar intrinsic-drift response is $\Xi(\lambda; h)$.

d. Scalar isotropic reduction.- We use Eq. (30) as the general threshold condition. We now extract a closed-form consequence in the common isotropic case. Suppose that the order parameter subspace is an irreducible representation of the transitive symmetry group of M , that the drift ensemble h is invariant under this symmetry, and that the phase-lag generator A preserves the same subspace. Then $\mathcal{S}(\lambda; h)$ commutes with the symmetry action and Schur's lemma gives

$$\mathcal{S}(\lambda; h)|_{\mathcal{E}} = s_M(\lambda; h) \mathbf{I}_{\mathcal{E}}, \quad (31)$$

on the order parameter subspace $\mathcal{E}$. We then determine the loss of linear stability from the scalar mode equations

$$1 = K s_M(\lambda; h) \beta_\ell, \quad \beta_\ell \in \text{spec}[(\mathbf{I} - \alpha A)|_{\mathcal{E}}]. \quad (32)$$

For a stationary loss of stability with $\alpha = 0$ and $s_M(0^+; h) > 0$, Eq. (32) reduces to the explicit threshold formula

$$K_c = \frac{1}{s_M(0^+; h)}. \quad (33)$$

If, in addition, the transport response factorizes on this subspace as $s_M(\lambda; h) = \kappa(M) \Xi(\lambda; h)$, then

$$K_c = \frac{1}{\kappa(M) \Xi(0^+; h)}. \quad (34)$$

Eq. (34) applies for $\alpha = 0$. In this factorized scalar reduction, K_c is inversely proportional to both the averaged tangent-projection factor $\kappa(M)$ and the drift susceptibility $\Xi(0^+; h)$. At fixed Ξ , increasing $\kappa(M)$ lowers the critical coupling required for the incoherent state to lose stability, while changes in the intrinsic heterogeneity modify $\Xi(0^+; h)$. Table III summarizes how the averaged tangent projection contributes to the coupling-induced source term and, after the intrinsic-drift response is evaluated, to the threshold condition.

e. Spherical reductions.- We first check the density-level formulation by recovering the standard Kuramoto threshold in the one-dimensional limit. For $M = S^1$, we write the two real embedding coordinates in complex notation as $F(\theta) = e^{i\theta}$, take $V = \omega \partial_\theta$, set $h(V) dV = g(\omega) d\omega$, and set $\alpha = 0$. The first Fourier mode of Eq. (28) gives the usual scalar self-consistency relation

$$1 = \frac{K}{2} \int_{\mathbb{R}} \frac{g(\omega)}{\lambda + i\omega} d\omega. \quad (35)$$

For an even unimodal frequency density and a stationary neutral mode, we take $\lambda \rightarrow 0^+$ and obtain

$$K_c = \frac{2}{\pi g(0)}, \quad (36)$$

with the conventional Kuramoto normalization [1–3]. This calculation recovers the full threshold formula in the circle limit and retains the intrinsic-drift distribution in the critical coupling.

For higher-dimensional hyperspheres, we keep D as the manifold dimension and write the standard case as $M = S^D \subset \mathbb{R}^{D+1}$. The same framework recovers the averaged tangent-projection coefficient $\kappa(S^D) = D/(D+1)$ in the coupling-induced source term. The threshold is then obtained from the susceptibility condition in Eq. (30) after one specifies the distribution of intrinsic rotation fields. We now specialize this condition to the standard hyperspherical rotation fields. Let $M = S^D \subset \mathbb{R}^{D+1}$, let $V_\Omega(\sigma) = \Omega\sigma$ with $\Omega \in \mathfrak{so}(D+1)$, and write the rotation-field distribution as $h(\Omega) d\Omega$. For $f_z(\sigma) = \sigma^\top z$, direct

spherical calculus gives

$$\begin{aligned}\nabla_{S^D} f_z &= \mathbf{P}_\sigma^\perp z, \\ \nabla_{S^D} \cdot (\mathbf{P}_\sigma^\perp z) &= -D f_z, \\ \mathcal{A}_{V_\Omega} f_z &= f_{\Omega z}.\end{aligned}\quad (37)$$

Thus the first spherical-harmonic subspace is invariant under the intrinsic transport, and Eq. (29) reduces to

$$\begin{aligned}\mathcal{S}_{S^D}(\lambda; h) z &= \frac{D}{D+1} \int_{\mathfrak{so}(D+1)} (\lambda \mathbf{I}_{D+1} - \Omega)^{-1} z \\ &\quad \times h(\Omega) d\Omega.\end{aligned}\quad (38)$$

Eq. (38) is the hyperspherical susceptibility associated with the generalized rotation field V_Ω . In this expression, the universal projection factor $D/(D+1)$ multiplies the drift resolvent, and the rotation-field distribution is included in the same resolvent average.

We obtain a scalar closed-form threshold by choosing a solvable rotation-field ensemble. Let $D+1 = 2m$, let J be a fixed complex structure with $J^\top = -J$ and $J^2 = -\mathbf{I}_{D+1}$, and choose $\Omega = \omega J$ with $h(\Omega) d\Omega = g(\omega) d\omega$. Then Eq. (38) gives

$$\mathcal{S}_{S^D}(\lambda; g) z = \frac{D}{D+1} \int_{\mathbb{R}} (\lambda \mathbf{I}_{D+1} - \omega J)^{-1} z g(\omega) d\omega. \quad (39)$$

If g is even, the term proportional to J cancels and

$$s_{S^D}(\lambda; g) = \frac{D}{D+1} \int_{\mathbb{R}} \frac{\lambda g(\omega)}{\lambda^2 + \omega^2} d\omega. \quad (40)$$

For an even unimodal density and a stationary neutral mode, $\lambda \rightarrow 0^+$ gives

$$K_c = \frac{D+1}{D\pi g(0)}, \quad D+1 = 2m. \quad (41)$$

The case $D=1$ reduces to Eq. (36). For $D>1$, Eq. (41) gives the threshold for this specified Hopf-rotation ensemble. Other rotation-field distributions are evaluated from Eq. (38). This solvable specialization displays how the averaged projection factor $D/(D+1)$ and the drift susceptibility enter the same threshold condition.

For the numerical check below, we write the scalar rotation-frequency density as g_Δ , where Δ is the width parameter of the intrinsic rotation-rate probability density. In the Lorentzian case, $g_\Delta(\omega) = \Delta/[\pi(\omega^2 + \Delta^2)]$, so $\pi g_\Delta(0) = 1/\Delta$ and Eq. (41) becomes

$$K_c^{\text{th}}(D, \Delta) = \frac{D+1}{D} \Delta. \quad (42)$$

For the Gaussian and uniform cases we use the same neutral formula $K_c^{\text{th}} = (D+1)/[D\pi g_\Delta(0)]$ with the corresponding value of $g_\Delta(0)$. We also evaluate the finite- λ susceptibility in Eq. (40) and define

$$K_c(\lambda; D, \Delta) = \frac{1}{s_{S^D}(\lambda; g_\Delta)}. \quad (43)$$

The limit $\lambda \rightarrow 0^+$ is the neutral spectral limit used to compute K_c . Thus Figure 1 tests the susceptibility calculation in Eq. (40) and the resulting formula for K_c . We next examine how a finite empirical sample of the geometric and intrinsic-drift distributions approaches the same calculation.

Figure 1 gives a numerical check of Eq. (41) on S^D . We use three probability densities to separate two effects: the dimension-dependent geometric factor $D/(D+1)$ and the drift response through $\pi g_\Delta(0)$. Figure 1(a) confirms that the susceptibility $s_{S^D}(\lambda; g_\Delta)$ has the required limit as $\lambda \rightarrow 0^+$. Figure 1(b) then tests the Lorentzian case, where Eq. (42) gives a straight-line relation between K_c^{th} and Δ . Figure 1(c) shows that the same formula for K_c also organizes Gaussian and uniform probability densities once each result is normalized by its own K_c^{th} . Therefore the numerics test the specific conclusion of the linearized kinetic equation: geometry determines the strength factor multiplying the coupling-induced source term, while the intrinsic drift distribution supplies the susceptibility factor in the critical coupling.

We further test the same threshold condition by replacing the density-level averages in Eq. (40) by finite empirical averages. For a sample $\{x_j, \omega_j\}_{j=1}^N$ on S^D , with the order parameter perturbation taken along a fixed embedding direction, we use

$$\kappa_N = \frac{1}{N} \sum_{j=1}^N (1 - x_{j,1}^2) \quad (44)$$

as the empirical tangent-projection factor. For the Lorentzian intrinsic rotation-rate density and a small spectral regularization $\epsilon = 10^{-2}\Delta$, we then compute

$$K_{c,N}^{(\epsilon)} = \left[\kappa_N \frac{1}{N} \sum_{j=1}^N \frac{\epsilon}{\epsilon^2 + \omega_j^2} \right]^{-1}. \quad (45)$$

The corresponding density-level quantity $K_c^{(\epsilon)}$ is obtained by replacing the empirical averages in Eq. (45) by their integrals. The comparison therefore checks the finite-sample convergence of the two factors in Eq. (45).

Figure 2 complements Figure 1. The first figure verifies the closed-form density-level susceptibility. The second figure verifies that the two empirical averages in Eq. (45), namely the tangent-projection average and the intrinsic-drift resolvent average, converge to the corresponding integral values. The numerical evidence therefore supports the interpretation that geometry fixes the coupling strength factor, whereas the intrinsic-drift probability density fixes the response factor that appears together with it in the critical coupling.

C. Non-linear response term of the order parameter response equation

In the previous section, we formulated the linearized kinetic spectral problem and identified the geometric

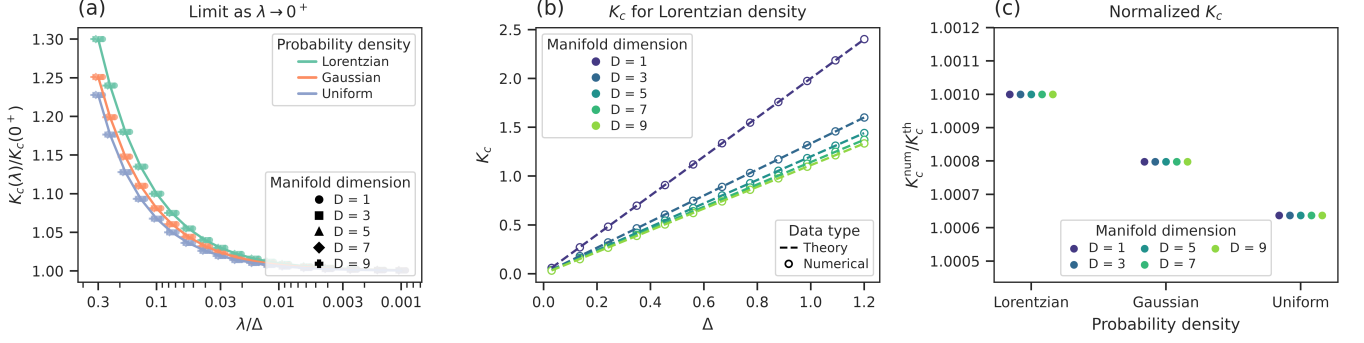


Figure. 1. Numerical verification of K_c on hyperspheres for the Hopf-rotation ensemble. The symbol D denotes the manifold dimension, so the tested manifolds are $S^D \subset \mathbb{R}^{D+1}$ with $D = 1, 3, 5, 7, 9$. The parameter Δ is the width parameter of the intrinsic rotation-rate probability density g_Δ , and $\lambda > 0$ is the parameter in Eq. (40) that is taken to 0^+ when computing K_c . (a) The plotted quantity is $K_c(\lambda; D, \Delta)/K_c(0^+; D, \Delta)$, where $K_c(\lambda; D, \Delta) = 1/s_{SD}(\lambda; g_\Delta)$. The ratio approaches one as $\lambda/\Delta \rightarrow 0^+$. (b) For the Lorentzian density, the computed value of $K_c(\lambda; D, \Delta)$ at $\lambda/\Delta = 10^{-3}$ follows the lines $K_c^{\text{th}} = (D+1)\Delta/D$. The different slopes show the effect of the geometric projection factor $D/(D+1)$. (c) After normalization by K_c^{th} , the computed values for Lorentzian, Gaussian, and uniform probability densities remain close to one for every tested manifold dimension. This shows that the geometric factor and the intrinsic-drift probability density enter the same formula for K_c .

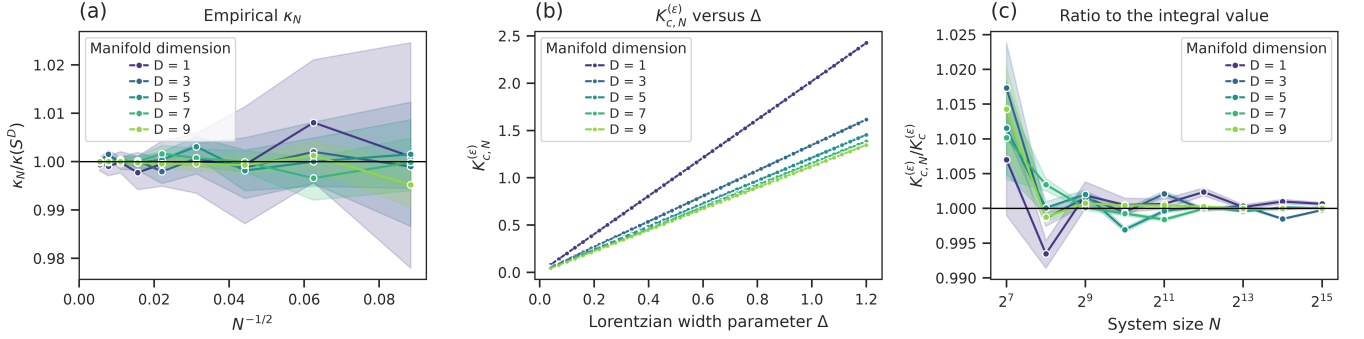


Figure. 2. Finite-sample verification of the critical coupling computed from Eq. (45). The symbol D denotes the manifold dimension, and the tested manifolds are $S^D \subset \mathbb{R}^{D+1}$ with $D = 1, 3, 5, 7, 9$. The intrinsic rotation-rate probability density is Lorentzian with width parameter Δ , and the spectral regularization is fixed at $\epsilon = 10^{-2}\Delta$. (a) The empirical geometric factor κ_N , normalized by the theoretical value $\kappa(S^D)$, converges to unity as the sample size increases. This verifies the finite-sample recovery of the averaged tangent projection. (b) For the largest sample size $N = 32768$, the empirical value $K_{c,N}^{(\epsilon)}$ follows the curves obtained by replacing the sums in Eq. (45) by integrals. The markers and bands show empirical seed averages and confidence intervals, and the thin dashed curves show the corresponding integral values. (c) The ratio $K_{c,N}^{(\epsilon)}/K_c^{(\epsilon)}$, averaged over all tested Δ values and random seeds, approaches one with increasing N . This confirms that the finite empirical approximation converges to the same calculation of K_c used in Figure 1.

source coefficient $\kappa(M)$ that multiplies the coupling-induced source term in the linearized kinetic equation [see Eqs. (25-30)]. We now focus on the nonlinear coefficient that determines the local bifurcation type near the incoherent state. We separate two levels of the argument. The Poincaré-Hopf index relation for the critical tangent field is topological and unconditional. The conclusions about the sign of the cubic coefficient and about the local phase transition scenario are conditional results of a one-mode local reduction. The technical reduction is collected in Appendix D; here we retain only the ingredients needed to state these conditions. We assume that a simple critical eigenvalue selects the critical collective mode. We also assume transversality, so this mode changes stability at a definite critical coupling. We further impose

the vanishing of the quadratic coefficient, which rules out a preferred sign of the order parameter at the incoherent state. Appendix D gives the precise form used here.

a. Reduced amplitude equation.- We assume that the steady response problem admits a smooth local reduction near the critical coupling $K = K_c$. We assume that the critical eigenvalue of the linearized order parameter operator is simple. We assume that this eigenvalue crosses zero transversely. We also assume that the quadratic normal-form coefficient vanishes. Appendix D then yields a one-dimensional scalar amplitude equation for the critical collective coordinate. This equation is the one-mode reduction of the full response equation $\frac{\partial}{\partial t}r = \mathcal{L}(r) + \mathcal{N}(r)$ in Eq. (17). Its linear term is the reduced action of $\mathcal{L}$ near $K = K_c$, whereas its cubic term is

Topological role	Detailed effect on the nonlinear response
Topology constrains the critical texture	We apply topology to the critical tangent field $u_c(\sigma) = P_\sigma^\perp(r_c - \alpha A r_c)$, not through the ambient order parameter vector alone. The zeros of u_c are defect cores, and Poincaré-Hopf gives $\sum_\ell \text{ind}_{p_\ell}(u_c) = \chi(M)$. If the zeros are simple, then $N_{\text{def}} \geq \chi(M) $. Thus a non-zero Euler characteristic imposes a non-zero net defect charge on the critical texture. This statement does not determine the sign of Λ_3 .
Topology can affect Λ_3 under stated assumptions	Under the locality assumptions of Appendix D and the signed dominance condition of Appendix E, the contribution of neighborhoods of the zeros weighted by their local indices controls the cubic coefficient. In that conditional class, the topological defect data can give $\Lambda_3 < 0$. Without these dynamical assumptions, $\chi(M) \neq 0$ alone is not a sign theorem for Λ_3 .
Topology can affect the local transition under stated assumptions	If the conditional sign analysis gives $\Lambda_3 < 0$ and the quintic coefficient satisfies $\Lambda_5 > 0$, the one-mode normal form has a stabilized subcritical branch and a discontinuous local transition. If $\chi(M) = 0$, the local branch is determined by the normal-form coefficients rather than by topology.

TABLE IV. Topological contribution to the nonlinear response analysis. The defect constraint is unconditional. The statements about Λ_3 and the local phase transition scenario require the locality condition of Appendix D, the sign condition of Appendix E, and, for a stabilized discontinuous branch, the additional condition $\Lambda_5 > 0$.

the leading nonlinear contribution generated by $\mathcal{N}$. The reduced equation reads

$$\Psi(a; K) = \lambda(K)a + \Lambda_3 a^3 + O(a^4) + O(|K - K_c|a^2) = 0, \quad (46)$$

together with the critical response branch $r(a) = a r_c + O(a^2)$, where r_c is the critical order parameter direction. The coefficient Λ_3 admits the explicit formula derived in Appendix D. This reduction does not change the model. It only separates the critical collective coordinate from the stable modes that are determined by it near threshold.

b. Topological relation.- We connect Λ_3 with the topology of M by associating r_c with the induced critical tangent field

$$u_c(\sigma) = P_\sigma^\perp(r_c - \alpha A r_c). \quad (47)$$

This field is the natural geometric object in our model. The microscopic transport law in Eq. (11) depends on the order parameter through the projected tangent field $P_\sigma^\perp(r - \alpha A r)$. For this reason, the topological information is carried by the induced critical tangent field u_c in the nonlinear response. It is not determined by r_c alone. Under the locality and symmetry conditions stated in Appendix D, the cubic coefficient can be written as

$$\Lambda_3 = \int_M T(\sigma)[u_c(\sigma), u_c(\sigma), u_c(\sigma)] d\mu(\sigma), \quad (48)$$

where $T(\sigma)$ is a smooth symmetric trilinear tensor field. We use these conditions to express the locality and symmetry structure of the model. The coupling term acts pointwise on the manifold, and the homogeneous structure organizes the relevant tensor fields into symmetry components. Appendix D states the condition under which the elimination of stable modes preserves this structure. The Poincaré-Hopf theorem [34] then implies that, when the Euler characteristic $\chi(M)$ of M

is non-zero, the field u_c must have zeros. This topologically forced zero set does not by itself determine the sign of Λ_3 . To obtain a definite sign statement, we impose the sign condition stated in Appendix E. This condition concerns the local cubic density $q(\sigma) = T(\sigma)[u_c(\sigma), u_c(\sigma), u_c(\sigma)]$. It requires the contributions from neighborhoods of the zeros to admit the indexed decomposition in Eq. (E3), and it requires the weighted indexed sum $\Gamma(u_c)$ to be positive and to dominate the remaining contribution that is not weighted by the indices, as in Eq. (E6). Within this conditional class, $\chi(M) \neq 0$ implies $\Lambda_3 < 0$. When $\chi(M) = 0$, the Poincaré-Hopf relation does not supply a sign for Λ_3 , and the sign is decided by the analytic structure of the reduced nonlinearity.

c. Bifurcation consequence.- We also use Appendix D to write $\lambda(K) = \lambda'(K_c)(K - K_c) + O((K - K_c)^2)$ with $\lambda'(K_c) \neq 0$, such that any nontrivial small-amplitude branch with $\Lambda_3 \neq 0$ satisfies

$$|a| \sim \sqrt{-\frac{\lambda(K)}{\Lambda_3}}. \quad (49)$$

Therefore, if $\lambda'(K_c) > 0$, then $\Lambda_3 > 0$ yields a supercritical continuous branch on the linearly unstable side, while $\Lambda_3 < 0$ yields a subcritical branch on the linearly stable side and therefore a discontinuous transition after the branch is stabilized by higher-order terms. We combine this analytic bifurcation criterion with the conditional sign statement from Appendix E. Thus, within the locality and sign assumptions of Appendices D and E, a non-zero Euler characteristic removes the supercritical small-amplitude synchronization branch from the one-mode local normal form. A zero Euler characteristic supplies no such sign restriction and leaves both continuous and discontinuous local scenarios to be decided by the analytic coefficients. The condition $\Lambda_3 \neq 0$ is also standard. This condition only states that the cubic term is the first nonlinear term that controls the lo-

cal branch. When symmetry leaves a higher-dimensional critical eigenspace, we apply the same local conclusions branch by branch after selecting a symmetry-broken critical direction. The present one-mode reduction should be read in that branchwise sense for such cases. Subsecs. IIID and IIIE, together with Appendices F-I, record the corresponding defect structure, the weak-field scaling laws near a continuous branch, and the quintic scales of the subcritical regime. We summarize the topological origin of the nonlinear response term in Table IV.

For the hypersphere $M = S^{D-1}$, we have $\chi(S^{D-1}) = 1 + (-1)^{D-1}$, so $\chi = 0$ precisely when D is even. The topological part of the present framework therefore identifies the same even/odd division that appears in high-dimensional Kuramoto models. The actual local branch on either side of this division is still determined by the conditional normal-form coefficients described above.

D. Critical defect structure of the critical mode

We now give the spatial interpretation of the unconditional topological constraint. The induced tangent field $u_c(\sigma) = P_\sigma^\perp(r_c - \alpha Ar_c)$ gives the local direction in which the synchronized component first develops on M . In the language of statistical physics, we regard u_c as the ordered texture selected at the threshold [43–45]. An isolated zero of u_c is then a topological defect of that texture, and the zero itself is the defect core. At such a point, the local ordered direction cannot be extended smoothly, because the critical mode does not select any tangent direction there.

We assume that all zeros of u_c are isolated. Then the Poincaré-Hopf theorem gives

$$\sum_{\ell=1}^{N_{\text{def}}} \text{ind}_{p_\ell}(u_c) = \chi(M), \quad (50)$$

where N_{def} denotes the number of defect cores in the critical texture. Under the generic nondegeneracy condition stated in Appendix H, each isolated zero has index ± 1 . Therefore

$$N_{\text{def}} \geq |\chi(M)|. \quad (51)$$

We identify the integer $\text{ind}_{p_\ell}(u_c)$ as the topological charge carried by the ℓ th defect core. Eq. (50) states that the net defect charge of the critical texture is fixed by the Euler characteristic of M .

This conclusion is independent of the sign of Λ_3 . A non-zero Euler characteristic prevents a smooth defect-free critical texture. In the generic case, it forces the ordered texture that first appears at threshold to contain at least $|\chi(M)|$ simple defect cores. The critical pattern therefore carries a finite background topological charge from the start.

When $\chi(M) = 0$, topology allows a smooth defect-free critical texture. In that sense, the Poincaré-Hopf

constraint no longer imposes a defect-count constraint. If critical defects are present and all of them are simple, then their positive and negative topological charges must balance. The critical texture may still contain local defects, but topology does not impose a nonzero background charge. Appendix H records this counting argument in a precise form.

E. Conditional local phase transition scenarios

We finally use the local reductions of Appendices F and G to classify the local statistical-physics scenario determined by the cubic and quintic coefficients near the loss of stability of the incoherent state. The important point is not that topology changes the numerical values of the generic mean-field exponents in this one-dimensional local theory. The important point is that, after the locality and sign conditions have been imposed, the Euler characteristic can restrict which normal-form branch is available.

If $\chi(M) \neq 0$ and the locality and sign conditions in Appendices D and E hold, then $\Lambda_3 < 0$. Within this conditional class, the generic supercritical continuous branch in Appendix F is absent from the one-mode normal form. The tricritical continuous branch discussed in Appendix G is also absent within the same class, because it would require $\Lambda_3 = 0$. Under the additional analytic condition $\Lambda_5 > 0$, the local transition is discontinuous, and the local hysteresis width and jump amplitude are then determined by Appendix G.

If $\chi(M) = 0$, topology imposes no sign constraint on Λ_3 . The continuous branch is obtained only when the analytic coefficient satisfies $\Lambda_3 > 0$. A discontinuous branch is obtained when $\Lambda_3 < 0 < \Lambda_5$. A tricritical continuous branch requires the additional tuning $\Lambda_3 = 0$ with $\Lambda_5 > 0$. Thus the zero Euler characteristic condition is nonselective in the local normal form: it removes the Poincaré-Hopf defect constraint, but the realized local scenario is still chosen by the analytic coefficients of the reduced equation. The local exponents are fixed by the reduced normal form itself.

We summarize these conditional local phase transition scenarios in Table V.

We next test the local consequence of the topological classification for the phase transition. The numerical experiment is performed on the reduced response equation derived above, with coefficients computed from explicit critical tangent fields and local cubic densities. The oscillator state spaces are S^D and $\mathbb{T}^D$, with $D = 1, \dots, 8$. For S^D , we use the projected constant field $u_c(x) = e_{D+1} - (e_{D+1} \cdot x)x$. For $\mathbb{T}^D$, we use $u_c(\theta) = (\sin \theta_1, \dots, \sin \theta_D)$. We compute the local defect indices from finite difference Jacobians at the zeros of these fields. We then construct local cubic densities, compute Λ_3 from the contribution of the zero neighborhoods weighted by their local indices and from the background contribution not weighted by the indices, and nu-

Euler characteristic	Additional coefficient condition	Local phase transition scenario	Leading local law
$\chi(M) \neq 0$	Appendices D and E give $\Lambda_3 < 0$; $\Lambda_5 > 0$ stabilizes the subcritical branch.	Conditional discontinuous transition with hysteresis.	$\Delta K_{\text{hys}} \sim \Lambda_3^2 / (4\lambda'(K_c) \Lambda_5)$, $ a_{\text{jump}} = \sqrt{-\Lambda_3/\Lambda_5}$
$\chi(M) = 0$	$\Lambda_3 > 0$	Continuous transition selected by the analytic normal-form coefficient, not by topology.	$\beta = \frac{1}{2}$, $\gamma = 1$, $\delta = 3$
$\chi(M) = 0$	$\Lambda_3 < 0 < \Lambda_5$	Discontinuous transition selected by the analytic normal-form coefficients, not by topology.	$\Delta K_{\text{hys}} \sim \Lambda_3^2 / (4\lambda'(K_c) \Lambda_5)$, $ a_{\text{jump}} = \sqrt{-\Lambda_3/\Lambda_5}$
$\chi(M) = 0$	$\Lambda_3 = 0 < \Lambda_5$	Tricritical continuous transition after analytic tuning by the normal-form coefficients.	$ a_* \sim K - K_c ^{1/4}$, $ a \sim h ^{1/5}$

TABLE V. Conditional local phase transition scenarios in the one-mode normal form. Topology alone supplies the Poincaré-Hopf defect constraint. The first row additionally uses the locality and sign conditions of Appendices D and E; the discontinuous branch also requires the stabilizing condition $\Lambda_5 > 0$. For $\chi(M) = 0$, the local branch is determined by the normal-form coefficients rather than by the Euler characteristic alone.

merically continue

$$\Psi(a; K) = \lambda'(K_c)(K - K_c)a + \Lambda_3 a^3 + \Lambda_5 a^5 + \Lambda_7 a^7 + \rho \lambda'(K_c)(K - K_c)a^3. \quad (52)$$

In the simulations shown in Figure 3, we set $\lambda'(K_c) = 1$. The transition type is assigned after the coefficients are computed and the response equation is continued.

Figure 3 tests the phase transition consequence of the indexed local-density calculation. Figure 3(a) shows how the computed sign of Λ_3 changes the local branch structure. Figure 3(b) verifies the numerical condition that converts the indexed contribution into the sign $\Lambda_3 < 0$: the background contribution not weighted by the indices remains smaller than the indexed contribution in the tested nonzero Euler characteristic samples. This panel therefore contains only the even-dimensional sphere samples, because they are the S^D cases with nonzero Euler characteristic in the $D = 1, \dots, 8$ test set. Figure 3(c) shows the resulting Λ_3 values. With nonzero Euler characteristic and the sign condition, the computed coefficients are negative; with zero Euler characteristic, both signs occur. Figure 3(d) then tests the quantitative prediction for the hysteresis width by comparing the value from Eq. (G15) with numerical continuation of Eq. (52). Figures 3(e) and 3(f) show that the same computed coefficients produce the transition classification over dimensions and in the (Λ_3, Λ_5) plane.

IV. REPRESENTATIVE GEOMETRIC AND TOPOLOGICAL INDICATORS

In this section, we apply the theory presented in Sec. III to representative homogeneous manifolds. We first identify the geometric source coefficient $\kappa(M)$ that is determined by the averaged tangent projection on the chosen embedding. This coefficient multiplies the coupling-induced source term in the linearized kinetic equation. Together with the intrinsic-drift response, it determines the critical coupling for the loss of stability of the incoherent state. We then state the Euler characteristic. This gives an unconditional prediction for the net defect charge of the critical texture through the Poincaré-Hopf theorem. When the locality conditions of Appendix D and the sign condition of Appendix E are also imposed, the same Euler characteristic gives the conditional cubic-sign information used in the one-mode normal form. Under those additional conditions, a non-zero Euler characteristic removes the generic continuous branch from the local normal form and can lead to a discontinuous transition when $\Lambda_5 > 0$, whereas for a zero Euler characteristic the local branch is determined by the normal-form coefficients. In Table VI, the first four cases have nonzero Euler characteristic and the last four have zero Euler characteristic. Appendix J collects the explicit geometric and topological computations.

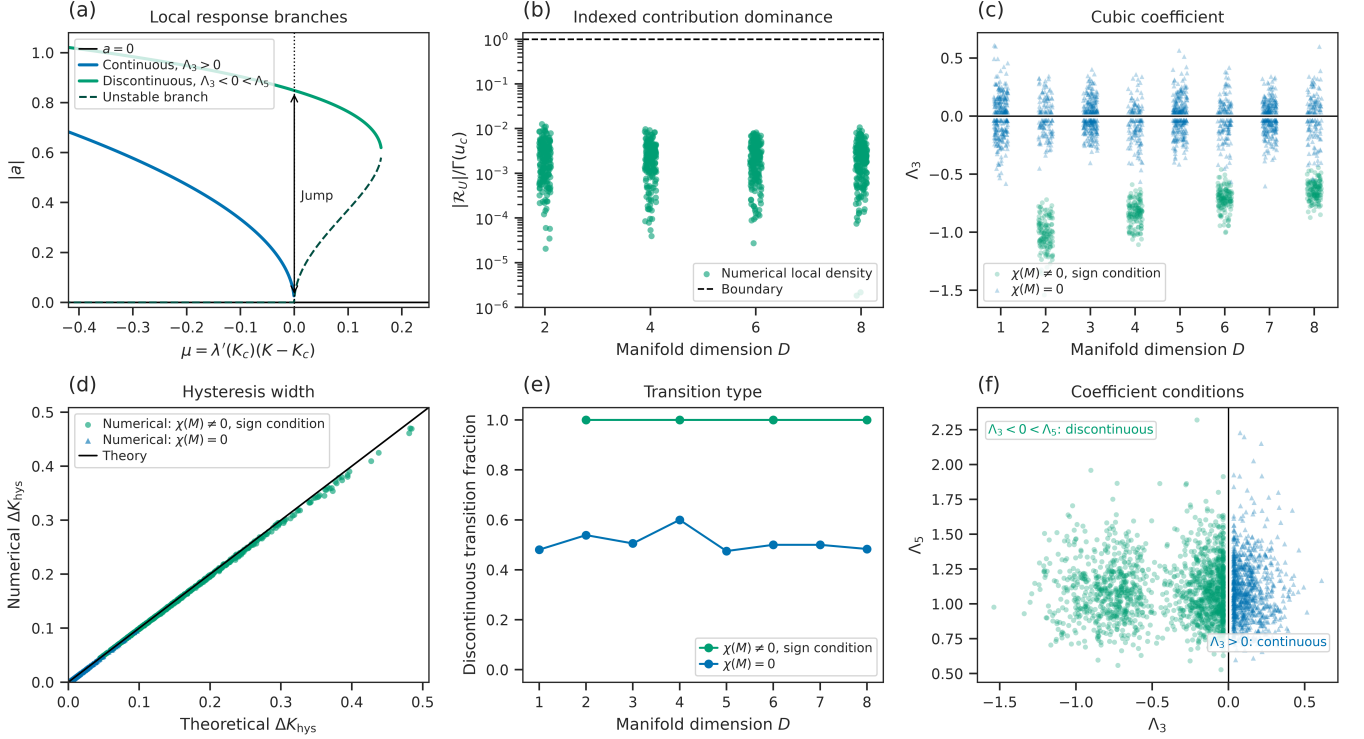


Figure 3. Numerical test of the topological contribution to the local transition type on S^D and $\mathbb{T}^D$. We write $\mu = \lambda'(K_c)(K - K_c)$, and the computed coefficients are inserted into Eq. (52). (a) Local response branches of the reduced equation. When $\Lambda_3 > 0$, the nonzero branch satisfies $a^2 = -\mu/\Lambda_3$, so the continuous branch exists for $\mu \leq 0$ and reaches $a = 0$ at $\mu = 0$. When $\Lambda_3 < 0 < \Lambda_5$, the finite-amplitude branch persists to the saddle-node point at positive μ , and its value at $\mu = 0$ gives the jump amplitude of the discontinuous transition. The dashed curve is the unstable branch. (b) Numerical evaluation of the sign condition for the nonzero Euler characteristic sphere cases. Only $D = 2, 4, 6, 8$ appear in this panel because $\chi(S^D) = 2$ for even D , whereas $\chi(S^D) = 0$ for odd D and $\chi(\mathbb{T}^D) = 0$ for all D . Here $\Gamma(u_c)$ is the contribution of the zero neighborhoods weighted by their local indices and $\mathcal{R}_U$ is the background contribution not weighted by the indices. The sampled ratios satisfy $|\mathcal{R}_U|/\Gamma(u_c) < 1$, which gives $\Lambda_3 = -\Gamma(u_c) + \mathcal{R}_U < 0$. (c) Cubic coefficient computed from the same local density. The sign condition in the nonzero Euler characteristic class gives $\Lambda_3 < 0$, whereas the zero Euler characteristic class contains both signs. (d) Hysteresis width of the local response equation. The horizontal coordinate is Eq. (G15), the vertical coordinate is obtained by numerical continuation of Eq. (52), and the black line is the theoretical diagonal. (e) Transition type over $D = 1, \dots, 8$. The class with $\chi(M) \neq 0$ and the sign condition gives discontinuous local transitions throughout the tested dimensions for which that class is present. The class with $\chi(M) = 0$ contains both transition types. (f) Classification in the (Λ_3, Λ_5) plane. The computed coefficients fall into the continuous condition $\Lambda_3 > 0$ or the discontinuous condition $\Lambda_3 < 0 < \Lambda_5$.

When the linearized problem is symmetry-degenerate, we interpret the one-mode reduction branch by branch. In other words, we first select a critical direction after symmetry breaking and then apply the local theory of Subsecs. IIIC-IIIE to that branch.

The entries in Table VI should be read as geometric and topological indicator data. The coefficient $\kappa(M)$ specifies how strongly the geometry of M enters the coupling-induced source term of the linearized kinetic equation, and it therefore directly affects the coupling strength required for the incoherent state to lose stability once the intrinsic-drift response has been fixed. The Euler characteristic gives the net topological charge that any critical texture must carry. The last column is conditional: it uses the local one-mode reduction only under the hypotheses stated in Appendices D and E.

We give the explicit averaged-projector, Euler-

characteristic, and defect-counting calculations in Appendix J. Appendix K provides a concise family-by-family interpretation of the entries in Table VI.

Figure 4 applies the same numerical procedure to non-spherical representative manifolds. For the geometric calculation, we represent $\mathbb{CP}^m$ and $\text{Gr}_2(\mathbb{C}^n)$ as Hermitian projector manifolds. We sample Haar-distributed projectors and isotropic traceless Hermitian ambient directions, project those directions onto the tangent space, and average the squared projection ratio to estimate $\kappa(M)$. We then insert the sampled $\kappa(M)$ into the Lorentzian intrinsic-drift response to compute K_c . For the topological calculation, we use the diagonal height-gradient field on the same projector manifolds. Its zeros are the coordinate projectors. We compute their local indices from the linearized tangent field in local charts, use the same local-density construction as in Figure 3 to obtain Λ_3 ,

Case	$\kappa(M)$	$\chi(M)$	Defect content	Conditional local scenario
S^{D-1}	$(D-1)/D$	$1 + (-1)^{D-1}$	Odd D : $Q_{\text{def}} = 2$, $N_{\text{def}}^{\text{min}} = 2$. Even D : $Q_{\text{def}} = 0$.	Odd D : under Appendices D–E, continuous and tricritical branches are absent; a discontinuous branch occurs if $\Lambda_5 > 0$. Even D : the Euler characteristic gives no branch selection.
$S^{2m} \times S^{2m}$	$\frac{2m}{2m+1}$	4	$Q_{\text{def}} = 4$, $N_{\text{def}}^{\text{min}} = 4$.	Under Appendices D–E, continuous and tricritical branches are absent. A discontinuous branch occurs if $\Lambda_5 > 0$.
$\text{Gr}_k(\mathbb{C}^n)$, $2 \leq k \leq n-2$	$\frac{2k(n-k)}{n^2-1}$	$\binom{n}{k}$	$Q_{\text{def}} = \binom{n}{k}$, $N_{\text{def}}^{\text{min}} = \binom{n}{k}$.	Under Appendices D–E, continuous and tricritical branches are absent. A discontinuous branch occurs if $\Lambda_5 > 0$.
$\mathbb{CP}^m$, $m \geq 2$	$\frac{2}{m+2}$	$m+1$	$Q_{\text{def}} = m+1$, $N_{\text{def}}^{\text{min}} = m+1$.	Under Appendices D–E, continuous and tricritical branches are absent. A discontinuous branch occurs if $\Lambda_5 > 0$.
$\mathbb{T}^d$	$1/2$	0	$Q_{\text{def}} = 0$. Defect-free texture allowed.	The Euler characteristic gives no branch selection. The normal-form coefficients decide the local scenario.
$\text{St}(p, n)$, $2 \leq p < n$	$(2n-p-1)/(2n)$	0	$Q_{\text{def}} = 0$. Defect-free texture allowed.	The Euler characteristic gives no branch selection. The normal-form coefficients decide the local scenario.
$\text{SO}(n)$	$(n-1)/(2n)$	0	$Q_{\text{def}} = 0$. Defect-free texture allowed.	The Euler characteristic gives no branch selection. The normal-form coefficients decide the local scenario.
$\text{U}(d)$	$1/2$	0	$Q_{\text{def}} = 0$. Defect-free texture allowed.	The Euler characteristic gives no branch selection. The normal-form coefficients decide the local scenario.

TABLE VI. Representative geometric and topological indicators. The first row is the hypersphere case; Rows 2–4 have $\chi(M) \neq 0$; Rows 5–8 have $\chi(M) = 0$. In Column 4, Q_{def} and $N_{\text{def}}^{\text{min}}$ denote the net defect charge and the minimum number of simple defect cores. Column 5 is stated within the one-mode reduction, under the locality conditions of Appendix D and the sign condition of Appendix E; a discontinuous branch is marked as selected only when the additional stabilizing condition $\Lambda_5 > 0$ holds. The status of these locality and sign assumptions for the listed families is summarized in Table VII.

and continue the response equation.

Figure 4(a) and Figure 4(b) compare the sampled averaged tangent projection with the analytic values of $\kappa(M)$ for two non-spherical families. Figure 4(c) shows how this geometric coefficient determines the critical coupling after the Lorentzian intrinsic-drift response is fixed. Figure 4(d) uses the diagonal height-gradient field to compute the index data on the same projector manifolds and then evaluates the corresponding local cubic coefficient. Figure 4(e) displays the phase transition type obtained from these computed coefficients. The families with nonzero Euler characteristic and the sign condition remain on the discontinuous branch, whereas the reference families with

zero Euler characteristic show both local transition types because the Euler characteristic does not fix the sign of Λ_3 . Figure 4(f) compares the hysteresis width from numerical continuation with Eq. (G15) for the samples with $\Lambda_3 < 0$ and $\Lambda_5 > 0$.

Finally, Figure 5 checks the phase transition scenarios by direct integration of finite oscillator systems. Figures 5(a)–5(c) show the hypersphere results. Figure 5(a) shows that the odd-dimensional zero Euler characteristic cases increase continuously in the forward scan, while the even-dimensional nonzero Euler characteristic cases show a finite jump. Figures 5(b) and 5(c) quantify the same distinction through the jump size and the maximum dif-

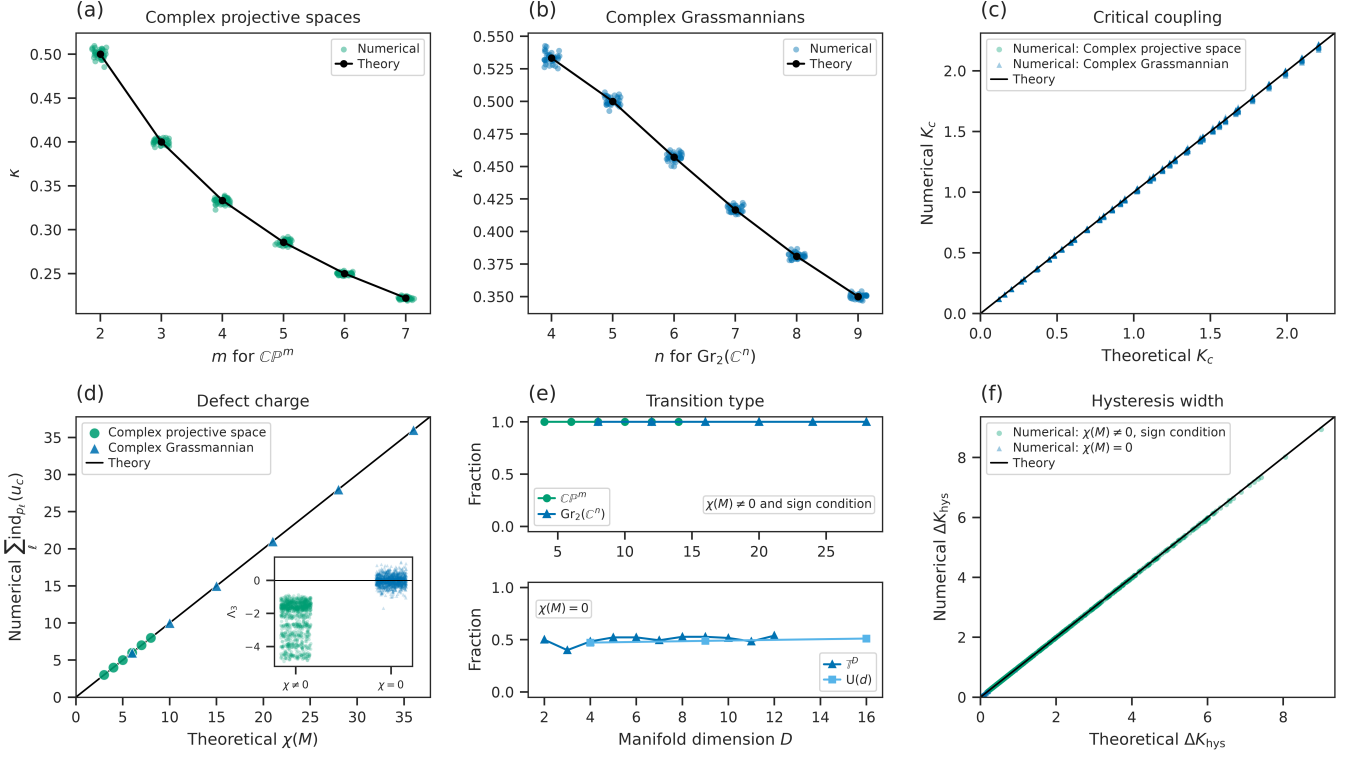


Figure. 4. Numerical validation on non-spherical homogeneous manifolds. The tested manifolds are complex projective spaces $\mathbb{CP}^m$ and complex Grassmannians $\text{Gr}_2(\mathbb{C}^n)$, represented by Hermitian projectors. (a) Monte Carlo estimate of the averaged tangent-projection factor on $\mathbb{CP}^m$. The numerical points follow the black curve $\kappa(\mathbb{CP}^m) = 2/(m+2)$. (b) Monte Carlo estimate of the averaged tangent-projection factor on $\text{Gr}_2(\mathbb{C}^n)$. The numerical points follow the black curve $\kappa(\text{Gr}_2(\mathbb{C}^n)) = 4(n-2)/(n^2-1)$. (c) Critical coupling computed with a Lorentzian intrinsic-drift distribution. The numerical value uses the sampled $\kappa(M)$, while the theoretical value uses the corresponding expression in Table VI. (d) Defect charge for the diagonal height-gradient field on the projector manifolds. The numerical index sum equals $\chi(M)$, and the inset shows the cubic coefficient computed from the corresponding local density. For nonzero Euler characteristic together with the sign condition, the computed coefficients satisfy $\Lambda_3 < 0$; for zero Euler characteristic, the topology does not impose a fixed sign. (e) Fraction of computed local response equations that give a discontinuous transition as the manifold dimension is varied. In the upper part, $\mathbb{CP}^m$ has real dimension $D = 2m$, and $\text{Gr}_2(\mathbb{C}^n)$ has real dimension $D = 4(n-2)$. These families with nonzero Euler characteristic and the sign condition select the discontinuous branch throughout the tested range. In the lower part, $\mathbb{T}^D$ has dimension D , and $\text{U}(d)$ has real dimension $D = d^2$. These reference families with zero Euler characteristic do not have a topologically fixed transition type. (f) Hysteresis width obtained by numerical continuation of the perturbed quintic response equation. The horizontal coordinate is Eq. (G15), the vertical coordinate is the numerical continuation result, and the black line is the theoretical diagonal.

ference between backward and forward scans.

Figure 5(d) extends the same finite- N test to product manifolds. The zero Euler characteristic controls $S^3 \times S^3$ and $S^3 \times S^5$ show continuous growth of the order parameter, whereas the sign-conditioned nonzero Euler characteristic cases $S^2 \times S^2$ and $S^2 \times S^4$ show finite jumps and large scan-direction differences, as quantified in Figures 5(e) and 5(f). Figures 5(g)–5(i) repeat the same test on non-spherical representatives from Table VI. The torus cases $\mathbb{T}^2$ and $\mathbb{T}^4$ provide zero Euler characteristic controls, while $\mathbb{CP}^2$ and $\mathbb{CP}^3$ provide nonzero Euler characteristic tests under a sign-conditioned Hermitian drift ensemble. The finite- N data therefore give a dynamical verification of the conditional topological prediction across hyperspheres, product manifolds, and non-spherical homogeneous manifolds.

V. CONCLUSION

In this work, we extended the Kuramoto-Sakaguchi framework from spheres to compact, connected, orientable, and homogeneous manifolds of arbitrary dimension. We derived the continuum equation on the manifold, identified the linear and nonlinear terms in the order parameter response equation near the incoherent state, and connected those terms with the geometry and topology of the state manifold. For the chosen embedding, the averaged tangent projection determines the coefficient $\kappa(M)$. This coefficient multiplies the coupling-induced source term in the linearized kinetic equation. The critical coupling K_c is determined by this geometric coefficient together with the intrinsic-drift response. The manifold topology affects the nonlinear response through

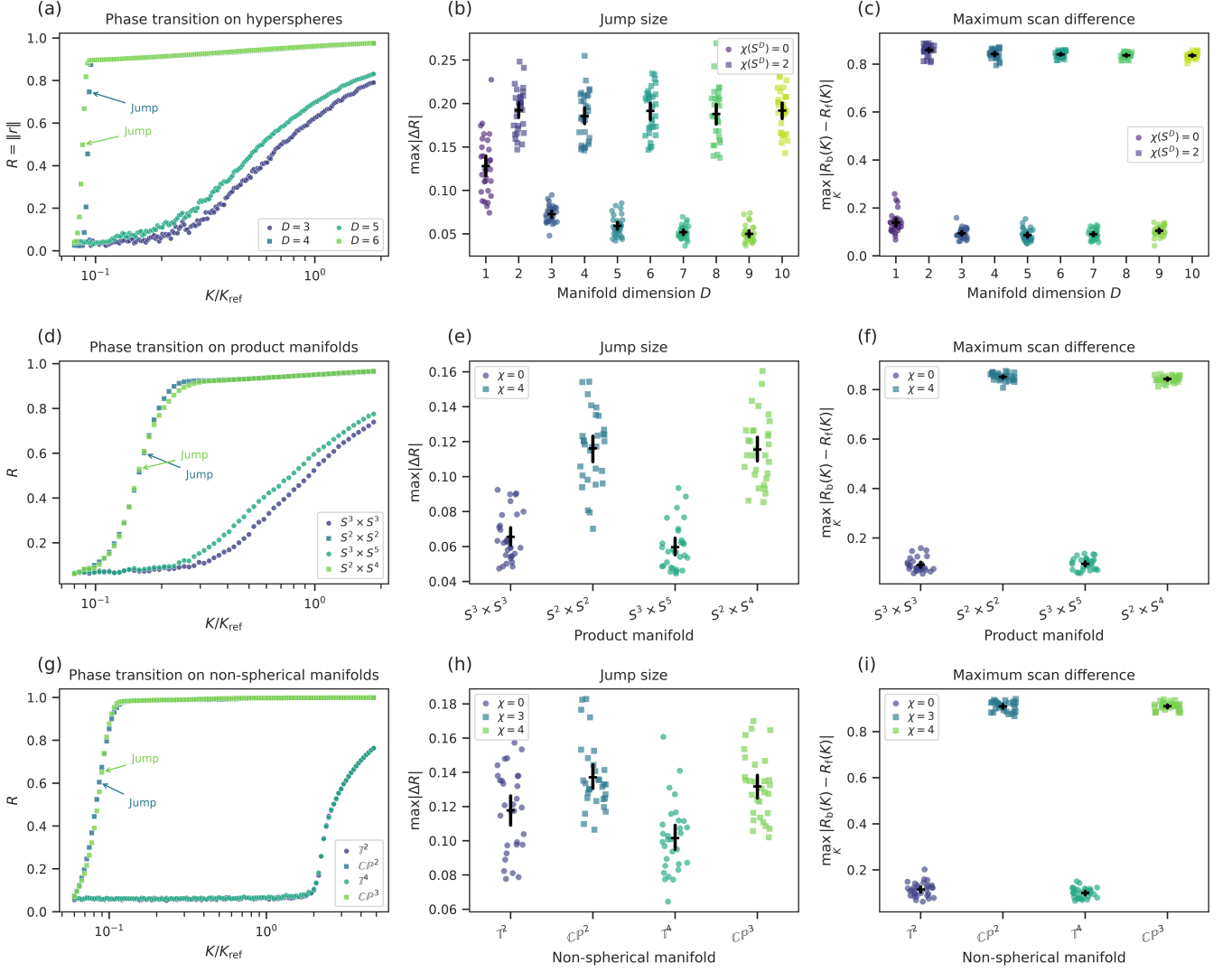


Figure. 5. Direct finite- N dynamical tests of the phase transition scenarios. The figure uses three groups of manifolds: hyperspheres S^D , product manifolds $S^p \times S^q$, and non-spherical representatives $\mathbb{T}^D$ and $\mathbb{CP}^m$. In all panels, R is the computed order parameter and $K_{\text{ref}} = \Delta$, where Δ is the width parameter of the intrinsic-drift distribution. The zero Euler characteristic cases use ordinary drift ensembles, whereas the nonzero Euler characteristic cases use structurally prescribed sign-conditioned drift ensembles. All generated samples are retained. (a) Forward K -scan for hyperspheres. The odd-dimensional cases have $\chi(S^D) = 0$ and show a continuous increase of R , whereas the even-dimensional cases have $\chi(S^D) = 2$ and show a finite jump. (b) Jump size $\max|\Delta R|$ over $D = 1, \dots, 10$. The square markers denote the even-dimensional nonzero Euler characteristic cases. (c) Maximum difference between the backward and forward scans, $\max_K |R_b(K) - R_f(K)|$, for the same hypersphere data. (d) Forward K -scan for product manifolds. The zero Euler characteristic controls $S^3 \times S^3$ and $S^3 \times S^5$ increase continuously, whereas the sign-conditioned nonzero Euler characteristic cases $S^2 \times S^2$ and $S^2 \times S^4$ show jumps. (e) Jump size for the product manifolds. (f) Maximum scan-direction difference for the product manifolds. (g) Forward K -scan for the non-spherical representatives $\mathbb{T}^2$, $\mathbb{T}^4$, $\mathbb{CP}^2$, and $\mathbb{CP}^3$. The torus cases have $\chi = 0$, whereas $\mathbb{CP}^2$ and $\mathbb{CP}^3$ have nonzero Euler characteristic and use the sign-conditioned Hermitian drift ensemble. (h) Jump size for these non-spherical representatives. (i) Maximum scan-direction difference for the same data. In the statistical panels, the colored markers are independent finite- N samples and the black markers show the sample mean with a 95% confidence interval.

the Euler characteristic. The unconditional consequence is the Poincaré-Hopf defect constraint on the critical tangent field. The phase transition consequence is conditional: within the one-mode local reduction of Appendices D and E, a nonzero Euler characteristic gives the cubic sign used in the reduced normal form, and the ad-

ditional stabilization condition $\Lambda_5 > 0$ gives a discontinuous local transition. When $\chi(M) = 0$, the Euler characteristic gives no sign selection for the cubic coefficient, and the local branch is determined by the normal-form coefficients.

We then evaluated these indicators on representative

homogeneous manifolds. The hypersphere case recovers the topological part of the classical parity distinction and shows how the odd and even cases fall on the two Euler-characteristic sides of the theory. The family $S^{2m} \times S^{2m}$ shows that the same nonzero Euler characteristic defect constraint extends beyond a single sphere while the geometric projection factor varies. The complex Grassmannian and complex projective families show that the nonzero Euler characteristic defect constraint also applies to non-spherical complex homogeneous spaces. The torus, real Stiefel, rotation-group, and unitary cases give the complementary zero Euler characteristic class: the Euler characteristic imposes no nonzero net defect charge, and the local branch is determined by the normal-form coefficients. These representative manifolds illustrate how the averaged tangent projection and the Euler characteristic enter the analysis: geometry fixes the geometric contribution to the coupling strength carried by the linear source term, while topology supplies an unconditional defect constraint and a conditional contribution to local branch selection. The numerical data and the experiment code used to generate our results are available in an open repository [46].

Several concrete statistical-physics directions now become accessible within the present framework:

1. Finite-size scaling near the loss of stability of the incoherent state. One can study how the finite- N order parameter fluctuations, Binder-like cumulants [47], and pseudocritical couplings depend on whether $\chi(M)$ vanishes. The most direct comparison is between odd hyperspheres or $\mathbb{CP}^m$, where the conditional normal form predicts a subcritical local branch under Appendices D and E, and zero Euler characteristic cases such as $\mathbb{T}^d$ or $U(d)$, where the branch is determined by the normal-form coefficients [48].
2. Noise-induced switching and metastability on manifolds with nonzero Euler characteristic. By adding intrinsic noise to the kinetic equation, one can derive the stationary order parameter distribution near the subcritical branch and compute switching times between incoherent and synchronized states. The present quintic normal form gives a direct starting point for studying how a conditionally selected discontinuous transition affects activation barriers and residence times [48].
3. Defect kinetics near the loss of stability of the incoherent state. The theory predicts a net defect charge and a lower bound on the number of simple defect cores when $\chi(M) \neq 0$. A natural next step is to simulate the critical texture on manifolds such as $S^{2m} \times S^{2m}$, $\mathbb{CP}^m$, and $\text{Gr}_k(\mathbb{C}^n)$, and then measure defect creation, motion, annihilation, and coarsening as K crosses the threshold [43–45].
4. Quenched disorder in the intrinsic drifts. The present theory already contains the drift field

through the measure $h(V)$. One can study how the width and symmetry of that disorder modify the critical coupling, the cubic coefficient, and the accessibility of the continuous branch. This is especially relevant for matrix-valued models on $\text{St}(p, n)$, $\text{SO}(n)$, and $U(d)$ [49–51].

5. Topology versus interaction-network structure. The present analysis is mean-field in character. A concrete next problem is to place the same manifold state spaces on sparse random graphs, modular graphs, or low-dimensional lattices, and ask whether the conditional local scenarios survive when the coupling network itself is no longer complete [41, 52–59].
6. Nonequilibrium hysteresis under ramps and external forcing. Appendix F and Appendix G give local static scaling and hysteresis scales. One can now study slow and fast parameter ramps, periodic driving, and weak external fields to determine how the loop area, jump point, and lag depend on the manifold topology. This direction is especially natural for the cases with nonzero Euler characteristic, where the conditional normal form can select the discontinuous side of the local bifurcation structure [7, 60–62].

ACKNOWLEDGEMENTS

Author appreciates Dr. Hedong Hou at the Institut de Mathématiques d’Orsay for their inspiring discussions. This project is supported by the High Performance Scientific Computing Research Program at Inplane Computing Technologies Ltd.

Appendix A: Gradient flow structure for $\alpha = 0$

In this appendix, we show that the coupling term in Eq. (3) admits a gradient-flow structure when the phase-lag parameter α vanishes.

When $\alpha = 0$, the coupling in Eq. (5) simplifies to

$$\Gamma(\sigma_i, \sigma_j; 0) = P_{\sigma_i}^\perp F(\sigma_j), \quad (\text{A1})$$

such that the discrete dynamics in Eq. (3) becomes

$$\frac{d}{dt}\sigma_i = V_i(\sigma_i) + \frac{K}{N} \sum_{j=1}^N P_{\sigma_i}^\perp F(\sigma_j). \quad (\text{A2})$$

Then, we construct a potential whose Riemannian gradient reproduces the coupling term in Eq. (A2). We consider the mean-field potential of the oscillator system

$$U(\sigma_1, \dots, \sigma_N) = -\frac{K}{2N} \sum_{i,j=1}^N \langle F(\sigma_i), F(\sigma_j) \rangle. \quad (\text{A3})$$

Under the standing assumption that the Riemannian metric on M is induced by the embedding $F : M \hookrightarrow \mathbb{R}^D$, the Riemannian gradient of U with respect to σ_i is given by

$$\nabla_{\sigma_i} U = -\frac{K}{N} \sum_{j=1}^N \mathbf{P}_{\sigma_i}^\perp F(\sigma_j) = -\frac{K}{N} \sum_{j=1}^N \Gamma(\sigma_i, \sigma_j; 0). \quad (\text{A4})$$

Hence, the dynamics can be written compactly as

$$\frac{d}{dt} \sigma_i = V_i(\sigma_i) - \nabla_{\sigma_i} U, \quad i = 1, \dots, N \quad (\text{A5})$$

when $\alpha = 0$. As described by this result, the coupling part of the manifold Kuramoto model is a gradient flow of the potential in Eq. (A3), perturbed by the intrinsic drift fields V_i . This generalizes the well-known gradient structure of the classic Kuramoto model on the circle and of its high-dimensional extensions on spheres or other related homogeneous manifolds [2, 3, 9–11, 63].

Appendix B: Special cases of the general model

In this appendix, we briefly explain how several classic synchronization models arise as the special cases of the general framework defined in Eqs. (3, 5). In all the cases, the state of oscillator i is denoted by $\sigma_i \in M$, the embedding is a smooth map $F : M \hookrightarrow \mathbb{R}^{D_a}$, and the coupling is defined by $\Gamma(\sigma_i, \sigma_j; \alpha) = \mathbf{P}_{\sigma_i}^\perp [(\mathbf{I} - \alpha A) F(\sigma_j)]$ with a fixed skew-symmetric matrix $A \in \mathfrak{so}(D_a)$ and a scalar phase-lag parameter α .

a. Kuramoto model on S^1 . We take the state manifold $M = S^1 \cong \mathbb{R}/2\pi\mathbb{Z}$. Each oscillator state $\sigma_i \in S^1$ is actually an angular coordinate $\sigma_i \in \mathbb{R}/2\pi\mathbb{Z}$. With this identification, we write the embedding as $F(\sigma_i) = (\cos \sigma_i, \sin \sigma_i)$, while the phase-lag is turned off by setting $A = \mathbf{0}$ and $\alpha = 0$. The tangent space at σ_i is one-dimensional and spanned by $JF(\sigma_i) = (-\sin \sigma_i, \cos \sigma_i)$, where J is the 90° counterclockwise rotation matrix $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. With these definitions, the projection $\mathbf{P}_{\sigma_i}^\perp$ extracts the component of $F(\sigma_j)$ along $JF(\sigma_i)$, such that Eq. (3) reduces to the classic Kuramoto model with coupling $\sin(\sigma_j - \sigma_i)$ [2, 3].

b. Kuramoto-Sakaguchi model on S^1 . We keep the same manifold $M = S^1$ and embedding $F(\sigma) = (\cos \sigma, \sin \sigma)$. Additionally, we introduce the phase frustration by choosing $A = J$ and a small nonzero α . The matrix J generates rotations in $\mathbb{R}^2$, and the factor $(\mathbf{I} - \alpha A)F(\sigma_j)$ may be viewed as a first-order approximation to the finite phase-lag action $\exp(-\alpha A)F(\sigma_j)$. Projecting onto the tangent direction at σ_i again yields a scalar coupling along $JF(\sigma_i)$, which has a form of $\sin(\sigma_j - \sigma_i) - \alpha \cos(\sigma_j - \sigma_i)$ (i.e., the first-order expansion of the Kuramoto-Sakaguchi interaction $\sin(\sigma_j - \sigma_i - \alpha)$ for small $|\alpha|$) [2, 3].

c. Generalized Kuramoto model on S^{D-1} . In this case, we define the state manifold as $M = S^{D-1} = \{x \in \mathbb{R}^D : \|x\| = 1\}$ and let the embedding be the canonical inclusion $F(\sigma) = \sigma$. Thus, each state σ_i is a unit vector in $\mathbb{R}^D$ and the tangent space at σ_i is $T_{\sigma_i} S^{D-1} = \{v \in \mathbb{R}^D : \langle v, \sigma_i \rangle = 0\}$. Given with $A = 0$ and $\alpha = 0$, the coupling term becomes $\Gamma(\sigma_i, \sigma_j; 0) = \mathbf{P}_{\sigma_i}^\perp \sigma_j = \sigma_j - \langle \sigma_i, \sigma_j \rangle \sigma_i$, which coincides with the standard high-dimensional coupling term on the unit sphere used in generalized Kuramoto models on S^{D-1} [9–11, 63].

d. Frustrated Kuramoto model on S^{D-1} . Again, we take $M = S^{D-1}$ and $F(\sigma) = \sigma$. Additionally, we allow a non-trivial phase-lag operator $A \in \mathfrak{so}(D)$ together with a small nonzero α . The factor $(\mathbf{I} - \alpha A)\sigma_j = \sigma_j - \alpha A\sigma_j$ is the first-order truncation of the corresponding finite phase-lag action generated by A , such that the interaction $\Gamma(\sigma_i, \sigma_j; \alpha)$ implements a multi-dimensional frustration of the Kuramoto coupling term on the sphere. Such matrix-valued phase lags generalize the scalar phase shift of the Kuramoto-Sakaguchi model and recover frustrated high-dimensional Kuramoto systems studied on S^{D-1} [15, 16].

e. Vector-phase Kuramoto model on $\mathbb{T}^d$. To represent systems defined with d angular degrees of freedom per oscillator, we let the state manifold be the flat torus $\mathbb{T}^d = (\mathbb{R}/2\pi\mathbb{Z})^d$ and write $\sigma_i = (\sigma_i^1, \dots, \sigma_i^d)$, where each $\sigma_i^k \in \mathbb{R}/2\pi\mathbb{Z}$. A natural embedding is obtained as the product of d circle embeddings

$$F(\sigma_i) = (\cos \sigma_i^1, \sin \sigma_i^1, \dots, \cos \sigma_i^d, \sin \sigma_i^d) \in \mathbb{R}^{2d}. \quad (\text{B1})$$

We define A to be block-diagonal with d copies of J on the diagonal, which yields a phase-lag operator that acts independently on each angular component. The projected coupling then gives a system of d coupled Kuramoto-type equations for the vector $(\sigma_i^1, \dots, \sigma_i^d)$, which can be interpreted as a vector-phase or multi-frequency Kuramoto model on $\mathbb{T}^d$.

f. Kuramoto models on Stiefel and rotation manifolds. For synchronization of matrices with orthogonality constraints, we define M as a matrix manifold such as the real Stiefel manifold [22] $\text{St}(p, n) = \{X \in \mathbb{R}^{n \times p} : X^\top X = \mathbf{I}_p\}$ or the special orthogonal group $\text{SO}(n)$ [64]. The embedding is the canonical matrix inclusion $F(X) = X$ into $\mathbb{R}^{n \times p}$ (or into $\mathbb{R}^{n \times n}$ for $\text{SO}(n)$). A direct choice of phase-lag operator is $A(X) = BX$ with $B \in \mathfrak{so}(n)$ fixed, such that $(\mathbf{I} - \alpha A)X_j = X_j - \alpha BX_j$ realizes a linearized matrix-valued phase shift along the group direction generated by B . The projection $\mathbf{P}_{\sigma_i}^\perp$ enforces the orthogonality constraint and Eq. (3) recovers Kuramoto-type consensus dynamics on Stiefel or rotation manifolds [12, 13, 39].

g. Lohe and unitary Kuramoto models. To describe synchronization of unitary matrices, we define M as a unitary matrix group (e.g., $\text{U}(d)$ or $\text{SU}(d)$ [64]) and embed each unitary $U \in \text{U}(d)$ as $F(U) = U$ in $\mathbb{C}^{d \times d} \cong \mathbb{R}^{2d^2}$. The Lie algebra $\mathfrak{u}(d)$ consists of skew-Hermitian matrices [64] and we let $A \in \mathfrak{u}(d)$ act on U by left or right

multiplication. The factor $(\mathbf{I} - \alpha A)U_j$ then implements a linearized phase-lag along the group direction generated by A . The projected coupling reproduces Lohe-type matrix models and quantum Kuramoto ensembles on unitary groups [14, 23–33].

h. Network-distributed Kuramoto models on M . Finally, the same manifold framework applies when oscillators are coupled through a network instead of a global mean field. In this case, the global scalar coupling strength K in Eq. (3) is replaced by a weighted adjacency matrix $K_{ij} \in \mathbb{R}$ and the mean-field sum is replaced by $\sum_{j=1}^N K_{ij} \Gamma(\sigma_i, \sigma_j; \alpha)$. For any of the choices of M , F and A described above, this definition creates the Kuramoto- or Kuramoto-Sakaguchi-type dynamics on complex networks, with the state manifold M encoding the internal symmetry and geometry of individual oscillators [2, 3, 40, 41, 51–62, 65–70].

Appendix C: Continuum limit and kinetic equation

In this appendix, we provide a formal derivation of the continuum limit for the general manifold-based Kuramoto-Sakaguchi model in Eqs. (3, 5). The strategy follows the standard mean-field limit for the classic Kuramoto model on S^1 and its generalizations [14, 42], which can be adapted to the present geometric setting.

To keep the derivation compact, we work in this appendix with the joint density on TM . The main-text notation is recovered by the factorization $\varrho(\sigma, V, t) = \rho(\sigma, V, t) h(V)$, where ρ is the conditional spatial density for a fixed intrinsic drift field and $h(V)$ is the time-independent distribution of V .

a. Empirical measure and weak formulation For a fixed number N of oscillators, the dynamics is defined by Eq. (3), which we recall here for convenience

$$\frac{d\sigma_i}{dt} = V(\sigma_i) + \frac{K}{N} \sum_{j=1}^N \Gamma(\sigma_i, \sigma_j; \alpha), \quad (\text{C1})$$

where $\sigma_i(t) \in M$ denotes the oscillator state. The coupling is

$$\Gamma(\sigma_i, \sigma_j; \alpha) = \mathbf{P}_{\sigma_i}^\perp [(\mathbf{I} - \alpha A) F(\sigma_j)], \quad (\text{C2})$$

where $F: M \hookrightarrow \mathbb{R}^{D_a}$ is a fixed smooth embedding, $\mathbf{P}_{\sigma_i}^\perp: \mathbb{R}^{D_a} \rightarrow T_{\sigma_i} M$ is the orthogonal projection, $A \in \mathfrak{so}(D_a)$ is a skew-symmetric matrix, and $\alpha \in \mathbb{R}$ is the phase-lag parameter.

The empirical measure on TM associated with the configuration $\{(\sigma_i(t), V_i)\}_{i=1}^N$ is defined by

$$\nu_t^N = \frac{1}{N} \sum_{i=1}^N \delta_{(\sigma_i(t), V_i)}, \quad (\text{C3})$$

where $\delta_{(\sigma, V)}$ is the Dirac mass on TM . By its construction, ν_t^N is a probability measure on TM . The corresponding configuration-space empirical measure on M is

the marginal $\mu_t^N = \frac{1}{N} \sum_{i=1}^N \delta_{\sigma_i(t)}$, which is obtained from ν_t^N by integrating out V .

To derive an evolution equation for ν_t^N , we consider it in a weak form. Given a smooth function $\Phi \in C^\infty(TM)$, we define

$$\begin{aligned} \langle \Phi, \nu_t^N \rangle &= \int_{TM} \Phi(\sigma, V) \nu_t^N(d\sigma dV) \\ &= \frac{1}{N} \sum_{i=1}^N \Phi(\sigma_i(t), V_i) \end{aligned} \quad (\text{C4})$$

Differentiating Eq. (C4) with respect to t and using the chain rule on the manifold M , we obtain

$$\frac{d}{dt} \langle \Phi, \nu_t^N \rangle = \frac{1}{N} \sum_{i=1}^N \left\langle \nabla_M \Phi(\sigma_i, V_i), \frac{d\sigma_i}{dt} \right\rangle_g, \quad (\text{C5})$$

where $\nabla_M \Phi$ denotes the Riemannian gradient of Φ and $\langle \cdot, \cdot \rangle_g$ is the Riemannian inner product on $T_\sigma M$ induced by the metric g .

Substituting Eq. (C1) into Eq. (C5) and separating intrinsic and coupling contributions gives

$$\begin{aligned} \frac{d}{dt} \langle \Phi, \nu_t^N \rangle &= \frac{1}{N} \sum_{i=1}^N \langle \nabla_M \Phi(\sigma_i, V_i), V(\sigma_i) \rangle_g \\ &+ \frac{K}{N^2} \sum_{i=1}^N \sum_{j=1}^N \langle \nabla_M \Phi(\sigma_i, V_i), \Gamma(\sigma_i, \sigma_j; \alpha) \rangle_g. \end{aligned} \quad (\text{C6})$$

Based on the empirical measure defined in Eq. (C3), the right side of Eq. (C6) can be written as

$$\begin{aligned} \frac{d}{dt} \langle \Phi, \nu_t^N \rangle &= \int_{TM} \langle \nabla_M \Phi(\sigma, V), V(\sigma) \rangle_g \nu_t^N(d\sigma dV) \\ &+ K \int_{TM} \int_{TM} \langle \nabla_M \Phi(\sigma, V), \Gamma(\sigma, \sigma'; \alpha) \rangle_g \\ &\quad \nu_t^N(d\sigma dV) \nu_t^N(d\sigma' dV'). \end{aligned} \quad (\text{C7})$$

Eq. (C7) is the weak formulation of the finite- N dynamics in terms of the empirical measure ν_t^N .

b. Mean-field limit and continuity equation We now formally take the mean-field limit $N \rightarrow \infty$. We assume that the empirical measures ν_t^N converge weakly to a Borel probability measure, i.e. ν_t on M for each $t \geq 0$

$$\nu_t^N \xrightarrow[N \rightarrow \infty]{\text{weakly}} \nu_t. \quad (\text{C8})$$

Eq. (C8) holds in the sense that $\langle \Phi, \nu_t^N \rangle \rightarrow \langle \Phi, \nu_t \rangle$ for all $\Phi \in C^\infty(TM)$. Based on Eq. (C8), we can formally obtain the limit in Eq. (C7)

$$\begin{aligned} \frac{d}{dt} \langle \Phi, \nu_t \rangle &= \int_{TM} \langle \nabla_M \Phi(\sigma, V), V(\sigma) \rangle_g \nu_t(d\sigma dV) \\ &+ K \int_{TM} \int_{TM} \langle \nabla_M \Phi(\sigma, V), \Gamma(\sigma, \sigma'; \alpha) \rangle_g \\ &\quad \nu_t(d\sigma dV) \nu_t(d\sigma' dV'). \end{aligned} \quad (\text{C9})$$

This is the weak formulation of the mean-field equation for the limit measure μ_t .

On the tangent bundle TM , it is natural to assume that ν_t admits a density $\rho(\sigma, V, t)$ with respect to the product of the Riemannian volume measure $\mu(d\sigma)$ on M and a reference measure dV on each fiber $T_\sigma M$, i.e.,

$$\nu_t(d\sigma dV) = \rho(\sigma, V, t) \mu(d\sigma) dV, \quad (C10)$$

with the normalization

$$\int_{TM} \rho(\sigma, V, t) \mu(d\sigma) dV = 1. \quad (C11)$$

In terms of this density, the continuum order parameter takes the form

$$r(t) = \int_{TM} \int_M F(\sigma') \rho(\sigma', V', t) d\mu(\sigma') dV'. \quad (C12)$$

Using the explicit form of the coupling in Eq. (C2), the interaction contribution in Eq. (C9) can be re-written as

$$\begin{aligned} & \int_{TM} \int_{TM} \langle \nabla_M \Phi(\sigma, V), \Gamma(\sigma, \sigma'; \alpha) \rangle_g \nu_t(d\sigma dV) \nu_t(d\sigma' dV') \\ &= \int_{TM} \left\langle \nabla_M \Phi(\sigma, V), \int_{TM} \Gamma(\sigma, \sigma'; \alpha) \nu_t(d\sigma' dV') \right\rangle_g \\ &= \int_{TM} \left\langle \nabla_M \Phi(\sigma, V), \int_{TM} \Gamma(\sigma, \sigma'; \alpha) \rho(\sigma', V', t) d\mu(\sigma') dV' \right\rangle_g d\nu_t(\sigma, V), \end{aligned} \quad (C13)$$

where we have used Eq. (C10) in the last line.

By substituting Eq. (C2) into the inner integral in Eq.

(C13) and using the linearity of $P_\sigma^\perp$ and $(\mathbf{I} - \alpha A)$ with respect to $F(\sigma')$, we obtain

$$\begin{aligned} \int_{TM} \Gamma(\sigma, \sigma'; \alpha) \rho(\sigma', V', t) d\mu(\sigma') dV' &= \int_{TM} P_\sigma^\perp [(\mathbf{I} - \alpha A) F(\sigma')] \rho(\sigma', V', t) d\mu(\sigma') dV' \\ &= P_\sigma^\perp \left[(\mathbf{I} - \alpha A) \int_{TM} F(\sigma') \rho(\sigma', V', t) d\mu(\sigma') dV' \right] \\ &= P_\sigma^\perp (r(t) - \alpha A r(t)), \end{aligned} \quad (C14)$$

where we have used the definition of the order parameter in Eq. (C12) in the last step. This shows explicitly that, in the mean-field limit, the averaged coupling can be expressed in terms of the order parameter as the projected field $P_\sigma^\perp (r(t) - \alpha A r(t))$, which is precisely the coupling structure used in Eq. (11) in our main text.

After substituting Eq. (C10) and Eqs. (C13-C14) into Eq. (C9) and integrating by parts on M with respect to the Riemannian volume $\mu(d\sigma)$ (based on that M is compact and without boundary), we obtain the weak formulation

$$\begin{aligned} & \frac{d}{dt} \int_{TM} \Phi(\sigma, V) \rho(\sigma, V, t) d\mu(\sigma) dV \\ &= - \int_{TM} \Phi(\sigma, V) \nabla_M \cdot (\rho(\sigma, V, t) \mathbf{v}(\sigma, V, t)) d\mu(\sigma) dV, \end{aligned} \quad (C15)$$

where the mean-field velocity field on TM is given by

$$\mathbf{v}(\sigma, V, t) = V(\sigma) + K P_\sigma^\perp (r(t) - \alpha A r(t)) \quad (C16)$$

according to our analysis of the averaged coupling.

Because Eq. (C15) holds for all smooth functions $\Phi \in C^\infty(TM)$, it is equivalent to the continuity equation on TM

$$\frac{\partial}{\partial t} \rho(\sigma, V, t) + \nabla_M \cdot (\rho(\sigma, V, t) \mathbf{v}(\sigma, V, t)) = 0, \quad (C17)$$

where $\mathbf{v}(\sigma, V, t)$ is given by Eq. (C16). Eq. (C17) serves as the kinetic equation underlying the macroscopic model Eqs. (12-14). The associated normalization $\int_{TM} \rho(\sigma, V, t) d\mu(\sigma) dV = 1$ is preserved for all t .

Appendix D: Local bifurcation reduction of the order parameter response equation

In this appendix, we provide the technical reduction used in Subsec. III C. This appendix has two purposes. The first purpose is to derive the scalar amplitude equation near the incoherent state. The second purpose is to state explicit conditions under which the cubic coefficient becomes a local integral over M . We assume that the steady order parameter response near the incoherent state is described by the equation

$$G(K, r) = \mathcal{L}(K) r + \mathcal{N}(K, r) = 0, \quad (\text{D1})$$

on a Banach space X of admissible order parameter perturbations, with $G \in C^4$ in a neighborhood of $(K_c, 0)$ and $G(K, 0) = 0$. Let

$$L_c = D_r G(K_c, 0), \quad (\text{D2})$$

and assume that L_c is Fredholm of index 0. We assume that $\ker L_c = \text{span}\{r_c\}$ is one-dimensional. We assume that the adjoint kernel is generated by $r_c^\dagger$. We normalize the dual pairing by $\langle r_c^\dagger, r_c \rangle = 1$. We also assume that the corresponding simple eigenvalue satisfies the transversality condition

$$\lambda(K_c) = 0, \quad \lambda'(K_c) \neq 0. \quad (\text{D3})$$

These assumptions describe a standard codimension-one instability. The critical mode r_c is the only neutral collective direction at K_c . The remaining directions are determined by this mode near threshold. The transversality condition states that the instability is crossed at a nonzero rate when K passes through K_c . We introduce the spectral projectors $P[f] = r_c \langle r_c^\dagger, f \rangle$ and $Q = \mathbf{I} - P$, and decompose

$$r = a r_c + w, \quad \langle r_c^\dagger, w \rangle = 0. \quad (\text{D4})$$

Expanding the nonlinear part at $K = K_c$ gives

$$\mathcal{N}(K_c, r) = \mathcal{N}_2(r, r) + \mathcal{N}_3(r, r, r) + O(\|r\|_X^4), \quad (\text{D5})$$

where $\mathcal{N}_2$ and $\mathcal{N}_3$ are the symmetric multilinear Taylor coefficients. To exclude a transcritical-type correction in the scalar normal form, we further impose the condition

$$\langle r_c^\dagger, \mathcal{N}_2(r_c, r_c) \rangle = 0. \quad (\text{D6})$$

Eq. (D6) is automatic whenever the steady response equation is odd under $r \mapsto -r$. In more general settings, we impose Eq. (D6) as an additional condition. Eq. (D6) states that the incoherent state has no preferred sign along the critical mode at quadratic order.

We give a concrete example in which Eq. (D6) follows from symmetry. Take the classical circle case $M = S^1$, write $F(\theta) = e^{i\theta}$, set $\alpha = 0$, and take $V = \omega \partial_\theta$ with an even frequency density $g(\omega)$. The half-period shift

$\theta \mapsto \theta + \pi$ preserves the volume measure and the intrinsic transport, while it sends $F(\theta)$ to $-F(\theta)$. Therefore a steady density with order parameter r is mapped to a steady density with order parameter $-r$. The corresponding steady response map satisfies

$$G(K, -r) = -G(K, r) \quad (\text{D7})$$

on the first Fourier mode. Expanding Eq. (D7) in the critical coordinate eliminates all even scalar terms in the kernel equation. In particular, the projected quadratic coefficient in Eq. (D6) vanishes. This example shows the mechanism behind the symmetry condition; outside such an explicit sign-reversing equivariance, Eq. (D6) must be verified or assumed for the chosen model. The range equation then yields

$$\begin{aligned} w(a; K) &= a^2 w_2 + O(a^3) + O(|K - K_c| a), \\ w_2 &= -(QL_c Q)^{-1} Q \mathcal{N}_2(r_c, r_c). \end{aligned} \quad (\text{D8})$$

The correction w_2 is the stable component generated by the quadratic self-interaction of the critical mode. Eq. (D8) shows that w_2 is not an independent degree of freedom. The stable part is determined by the critical amplitude. Eq. (D8) then enters the kernel equation and yields the reduced amplitude equation

$$\begin{aligned} \Psi(a; K) &= \lambda(K) a + \Lambda_3 a^3 \\ &+ O(a^4) + O(|K - K_c| a^2) = 0, \end{aligned} \quad (\text{D9})$$

with

$$\Lambda_3 = \langle r_c^\dagger, \mathcal{N}_3(r_c, r_c, r_c) + 2\mathcal{N}_2(r_c, w_2) \rangle. \quad (\text{D10})$$

Eq. (D9) is the technical statement used in Subsec. III C. Eq. (D10) shows that the cubic coefficient contains a direct cubic term and a mixed term produced by the stable correction. Moreover, Eq. (D3) implies the local expansion

$$\lambda(K) = \lambda'(K_c)(K - K_c) + O((K - K_c)^2), \quad (\text{D11})$$

so every nontrivial small-amplitude branch with $\Lambda_3 \neq 0$ satisfies

$$|a| \sim \sqrt{-\frac{\lambda(K)}{\Lambda_3}} = \sqrt{-\frac{\lambda'(K_c)(K - K_c)}{\Lambda_3}}. \quad (\text{D12})$$

Eq. (D12) is the usual square-root scaling near a cubic bifurcation.

We next connect Eq. (D10) with the topology of M . We introduce the induced critical tangent field

$$u_c(\sigma) = P_\sigma^\perp(r_c - \alpha A r_c). \quad (\text{D13})$$

This field is the relevant geometric object for the nonlinear transport. The microscopic velocity in Eq. (11) depends on the order parameter through the projected

field $P_\sigma^\perp(r - \alpha Ar)$. For this reason, the topological content of the critical response is carried by u_c . For any stable correction $w \in QX$, we also define the induced tangent field

$$\tilde{u}_w(\sigma) = P_\sigma^\perp(w - \alpha Aw). \quad (\text{D14})$$

We now state a sufficient set of assumptions under which the cubic coefficient becomes pointwise local. These assumptions are not consequences of the Poincaré-Hopf theorem, and they are not automatic consequences of homogeneity. They are hypotheses on the nonlinear response operator and on the stable inverse in Eq. (D8). In particular, the correction w_2 contains the inverse of the global operator QL_cQ , so its locality must be checked for the chosen model and critical subspace.

a. Locality assumptions.— Let $\mathcal{S}_2(u_c) \subset QX$ denote the set of stable corrections w with the following property. The induced tangent field $\tilde{u}_w(\sigma)$ can be written pointwise as $S(\sigma)[u_c(\sigma), u_c(\sigma)]$ for some $S \in \Gamma(\text{Sym}^2 T^*M \otimes TM)$. This set collects the local quadratic tensor subspace generated by the critical field. It is a specified subspace of stable corrections, not the full stable space.

Assumption D1: local direct cubic term. We first require the direct cubic contribution to be local in the induced tangent field. This requirement states that the direct cubic response is assembled from a pointwise cubic density on M . Concretely, we assume that there exists a smooth symmetric trilinear tensor field $T_{3,0}(\sigma)$ such that

$$\begin{aligned} & \langle r_c^\dagger, \mathcal{N}_3(r_c, r_c, r_c) \rangle \\ &= \int_M T_{3,0}(\sigma)[u_c(\sigma), u_c(\sigma), u_c(\sigma)] d\mu(\sigma), \end{aligned} \quad (\text{D15})$$

Assumption D2: closure of the quadratic stable correction. We next require the quadratic source to stay inside the local quadratic subspace. We also require the stable inverse to preserve that subspace:

$$Q\mathcal{N}_2(r_c, r_c) \in \mathcal{S}_2(u_c). \quad (\text{D16})$$

$$(QL_cQ)^{-1}\mathcal{S}_2(u_c) \subseteq \mathcal{S}_2(u_c). \quad (\text{D17})$$

The second inclusion is the main locality hypothesis. It says that solving the stable range equation does not generate nonlocal dependence on the critical field inside the selected subspace. Under Eq. (D16) and Eq. (D17), the stable correction produced by the quadratic self-interaction remains local and quadratic in u_c . Therefore there exists a smooth symmetric bilinear bundle map $S_2(\sigma)$. This map satisfies

$$\tilde{u}_{w_2}(\sigma) = S_2(\sigma)[u_c(\sigma), u_c(\sigma)]. \quad (\text{D18})$$

Assumption D3: local mixed term. We finally require the mixed quadratic contribution to be local in the pair

$(u_c, \tilde{u}_w)$. This requirement states that the interaction between the critical mode and the stable correction is evaluated pointwise on M after the stable correction has been restricted to $\mathcal{S}_2(u_c)$. We therefore assume that there exists a smooth bilinear form $B(\sigma)$. This form satisfies the following relation for every $w \in \mathcal{S}_2(u_c)$:

$$\langle r_c^\dagger, \mathcal{N}_2(r_c, w) \rangle = \int_M B(\sigma)[u_c(\sigma), \tilde{u}_w(\sigma)] d\mu(\sigma). \quad (\text{D19})$$

Eq. (D18) turns Eq. (D19) into

$$\langle r_c^\dagger, \mathcal{N}_2(r_c, w_2) \rangle = \int_M T_{3,1}(\sigma)[u_c(\sigma), u_c(\sigma), u_c(\sigma)] d\mu(\sigma), \quad (\text{D20})$$

The tensor field $T_{3,1}(\sigma)$ is obtained by symmetrizing the contraction of $B(\sigma)$ with $S_2(\sigma)$. Eqs. (D10, D15, and D20) then give

$$\Lambda_3 = \int_M T(\sigma)[u_c(\sigma), u_c(\sigma), u_c(\sigma)] d\mu(\sigma), \quad (\text{D21})$$

with $T = T_{3,0} + 2T_{3,1}$. This is precisely the local tensor representation used in Subsec. III C. The three assumptions above state that the direct cubic term, the quadratic correction, and their mixed contribution all preserve the local tensor structure selected by the projected tangent field.

A useful way to verify Eq. (D17) is through symmetry, but the required closure must still be checked. Suppose that the quadratic source generated by $u_c \otimes u_c$ belongs to a finite-dimensional G -invariant harmonic subspace for the transitive isometry group G of M , and suppose that QL_cQ is G -equivariant and invertible on that same subspace. If the subspace is irreducible, Schur's lemma [64] implies that QL_cQ acts there as a scalar multiple of the identity. This verifies the preservation condition in Eq. (D17) on that subspace. Without such a componentwise verification, Eq. (D21) should be read as an additional local reduction assumption rather than as a theorem following from homogeneity. Under Assumptions D1–D3, Subsec. III C reduces the cubic coefficient to the topology of the induced tangent field. Appendix E then supplies a separate sign condition needed to pass from the forced zeros of u_c to a definite sign of Λ_3 .

Appendix E: Sign condition for the cubic coefficient

In this appendix, we state the additional sign condition used in Subsec. III C. Under the locality assumptions of Appendix D, the cubic coefficient can be written as a local integral over M . The Poincaré-Hopf theorem fixes the total index of the zeros of u_c , but it does not by itself fix the sign of this integral. The sign statement therefore requires an additional model-dependent dominance condition on the local cubic density. We write this density

in Eq. (48) as

$$q(\sigma) = T(\sigma) [u_c(\sigma), u_c(\sigma), u_c(\sigma)], \quad (\text{E1})$$

The density $q(\sigma)$ measures the local contribution of the critical tangent field to the cubic coefficient. The coefficient Λ_3 is the integral of this density:

$$\Lambda_3 = \int_M q(\sigma) d\mu(\sigma). \quad (\text{E2})$$

We assume that the zero set of u_c consists of isolated points $\{p_1, \dots, p_m\}$. This situation is generic for a smooth vector field after a small perturbation. For each zero p_ℓ , we choose a small geodesic neighborhood U_ℓ [36]. The sets U_ℓ are disjoint and contain no other zeros. We decompose the local contribution from each core as

$$\int_{U_\ell} q(\sigma) d\mu(\sigma) = -\gamma_\ell \text{ind}_{p_\ell}(u_c) + \delta_\ell, \quad \gamma_\ell > 0. \quad (\text{E3})$$

Here γ_ℓ measures the indexed part of the local contribution, and δ_ℓ records the local remainder not weighted by the index. The sign convention in Eq. (E3) is the one used in the main text. We then define the weighted indexed sum

$$\Gamma(u_c) = \sum_{\ell=1}^m \gamma_\ell \text{ind}_{p_\ell}(u_c). \quad (\text{E4})$$

The Euler characteristic fixes the unweighted sum of the indices, not the weighted sum in Eq. (E4). Thus the condition $\chi(M) \neq 0$ does not by itself imply $\Gamma(u_c) > 0$ when the local indices have mixed signs or when the weights vary.

Let $U = \bigcup_{\ell=1}^m U_\ell$ and define the remainder not weighted by the indices by

$$\mathcal{R}_U = \sum_{\ell=1}^m \delta_\ell + \int_{M \setminus U} q(\sigma) d\mu(\sigma). \quad (\text{E5})$$

The sign condition used in the main text is the following signed dominance assumption:

$$\Gamma(u_c) > 0, \quad |\mathcal{R}_U| < \Gamma(u_c). \quad (\text{E6})$$

Under Eq. (E6), Eq. (E2) becomes

$$\Lambda_3 = -\Gamma(u_c) + \mathcal{R}_U < 0. \quad (\text{E7})$$

This is the additional sign condition used in the main text when we pass from the topological defect constraint to a definite subcritical sign of the cubic normal-form coefficient. The quantities γ_ℓ , δ_ℓ , and $\mathcal{R}_U$ depend on the nonlinear response operator, the chosen critical branch, and the drift ensemble. They are not determined by the Euler characteristic alone.

When $\chi(M) = 0$, the above sign conclusion is unavailable. In that case the sign of Λ_3 must be determined from the analytic structure of $T(\sigma)$ and from the detailed nonlinear reduction.

Appendix F: Dynamical normal form and weak-field scaling

In this appendix, we record two direct consequences of the reduction derived in Appendix D. The first consequence is a scalar dynamical normal form near the critical coupling. The second consequence is the standard weak-field scaling law near a continuous branch. These statements do not add a new topological mechanism. They use the same one-dimensional reduction that already underlies Subsec. III C.

a. Dynamical normal form.— We assume that the local order parameter dynamics near the incoherent state is generated by the same response map as in Appendix D. More precisely, we assume that

$$\frac{d}{dt}r = -G(K, r), \quad (\text{F1})$$

where $G(K, r)$ is defined by Eq. (D1). We also assume that Eq. (F1) admits a C^3 local center manifold [71] at $(K_c, 0)$. The center manifold is tangent to $\text{span}\{r_c\}$ and uses the same critical mode as the steady reduction in Appendix D. Therefore the reduced center variable a satisfies

$$\begin{aligned} \frac{d}{dt}a &= -\Psi(a; K) \\ &= -\lambda(K)a - \Lambda_3 a^3 \\ &\quad + O(a^4) + O(|K - K_c|a^2), \end{aligned} \quad (\text{F2})$$

where $\Psi(a; K)$ is the steady reduced equation in Eq. (D9). Hence the equilibria of Eq. (F2) are exactly the small-amplitude steady branches already studied in Subsec. III C.

Let $a_*(K)$ be a stable equilibrium of Eq. (F2). Linearizing Eq. (F2) around $a_*(K)$ gives the local relaxation rate

$$\tau^{-1}(K) = |\partial_a \Psi(a_*(K); K)|. \quad (\text{F3})$$

For the incoherent branch $a_* = 0$, Eq. (F3) gives

$$\tau^{-1}(K) = |\lambda(K)| + o(|K - K_c|), \quad (\text{F4})$$

and therefore

$$\tau(K) \sim |\lambda(K)|^{-1} \sim |K - K_c|^{-1}. \quad (\text{F5})$$

This is the usual critical slowing-down law. When a continuous branch exists, Eq. (D12) gives $a_*^2(K) = -\lambda(K)/\Lambda_3 + o(\lambda)$. Substituting this relation into Eq. (F3) yields

$$\begin{aligned} \tau^{-1}(K) &= |\lambda(K) + 3\Lambda_3 a_*^2(K)| \\ &= 2|\lambda(K)| + o(|K - K_c|), \end{aligned} \quad (\text{F6})$$

so the same scaling $\tau \sim |K - K_c|^{-1}$ holds on the continuous branch.

b. Weak-field scaling.- We next add a weak external bias in the critical direction. We assume that the full response equation takes the form

$$\frac{d}{dt}r = -G(K, r) + h\eta, \quad (\text{F7})$$

where $\eta \in X$ is a fixed bias direction with $\langle r_c^\dagger, \eta \rangle \neq 0$. After rescaling the field amplitude h , the reduced steady equation on the center manifold becomes

$$\begin{aligned} \Psi_h(a; K) &= \lambda(K)a + \Lambda_3 a^3 - h \\ &+ O(a^4) + O(|K - K_c|a^2) + O(|h||a|) \\ &= 0. \end{aligned} \quad (\text{F8})$$

At the critical coupling $K = K_c$, Eq. (F8) gives

$$|a| \sim \left| \frac{h}{\Lambda_3} \right|^{1/3}, \quad (\text{F9})$$

provided $\Lambda_3 \neq 0$. This is the standard mean-field field-response law, so the exponent is $\delta = 3$.

When $h = 0$ and a continuous branch exists, Eq. (D12) already gives

$$|a_*(K)| \sim |K - K_c|^{1/2}, \quad (\text{F10})$$

so the order parameter exponent is $\beta = 1/2$. To obtain the differential susceptibility, we differentiate Eq. (F8) with respect to h along a steady branch. This gives

$$\frac{\partial}{\partial h}a_* = (\lambda(K) + 3\Lambda_3 a_*^2(K))^{-1} + o(|K - K_c|^{-1}). \quad (\text{F11})$$

On the incoherent side, where $a_* = 0$, Eq. (F11) gives

$$\frac{\partial}{\partial h}a_* \sim \lambda(K)^{-1}. \quad (\text{F12})$$

On a continuous branch, Eqs. (D12, F11) give

$$\frac{\partial}{\partial h}a_* \sim (-2\lambda(K))^{-1}. \quad (\text{F13})$$

Therefore, on both sides of a continuous transition,

$$\left| \frac{\partial}{\partial h}a_* \right| \sim |K - K_c|^{-1}, \quad (\text{F14})$$

which is the standard mean-field susceptibility law $\gamma = 1$.

The present one-dimensional reduction therefore gives the usual mean-field triplet $\beta = \frac{1}{2}$, $\gamma = 1$, and $\delta = 3$ whenever a continuous branch exists. In the present theory, topology does not change these exponents at the generic codimension-one level. Instead, topology affects the availability of the continuous branch described by Eqs. (F9-F14). Under the sign condition in Appendix E, a manifold with $\chi(M) \neq 0$ gives $\Lambda_3 < 0$, so that branch is absent from the conditional one-mode normal form. A manifold with $\chi(M) = 0$ leaves the sign of Λ_3 to the analytic coefficients.

Appendix G: Quintic normal form and hysteresis scales

In this appendix, we extend the local reduction to quintic order. This extension is useful when the cubic coefficient is negative and the continuous branch is therefore absent in the conditional normal form. The cubic normal form then does not determine the stable large-amplitude branch by itself. The hysteresis analysis only needs one additional property beyond Appendix D. The reduced scalar equation must be odd up to quintic order:

$$\Psi(-a; K) = -\Psi(a; K) + O(a^7) + O(|K - K_c|a^3). \quad (\text{G1})$$

This condition is natural when the reduced order parameter has no preferred sign and no external bias is present. The amplitudes a and $-a$ describe the same critical mode with opposite sign. In the absence of an external field or built-in chirality, the reduced dynamics should not distinguish them.

A weaker sufficient condition at the model level is a local $\mathbb{Z}_2$ symmetry of the response map near the incoherent state. Assume that there exists a bounded linear involution $\mathcal{R} : X \rightarrow X$ such that

$$\mathcal{R}^2 = \mathbf{I}, \quad G(K, \mathcal{R}r) = \mathcal{R}G(K, r), \quad \mathcal{R}r_c = -r_c.$$

Then L_c commutes with $\mathcal{R}$. The local center manifold can be chosen $\mathcal{R}$ -invariant. The amplitude coordinate changes from a to $-a$. The reduced scalar equation is therefore odd. Eq. (G1) follows. This condition is weaker than $G(K, -r) = -G(K, r)$. This condition is also weaker than assuming a specific involutive isometry of M . A manifold isometry is only one concrete realization of this local $\mathbb{Z}_2$ symmetry.

If Eq. (G1) fails, the reduced scalar equation contains even terms. The two signs of a are then not equivalent. The local bifurcation becomes imperfect or asymmetric. The symmetric branch formulas derived below do not hold in the same form.

Eq. (G1) is enough for the quintic normal form and for the hysteresis scalings derived below. We now keep the general Taylor expansion of $\mathcal{N}(K_c, r)$ up to quintic order. We also assume that $G \in C^7$ in a neighborhood of $(K_c, 0)$, so the reduced equation admits a controlled quintic expansion. The range correction then has the form

$$\begin{aligned} w(a; K) &= a^2 w_2 + a^3 w_3 + a^4 w_4 \\ &+ O(a^5) + O(|K - K_c|a). \end{aligned} \quad (\text{G2})$$

Here w_2 is given by Eq. (D8). The coefficients $w_3, w_4 \in QX$ satisfy

$$w_3 = -(QL_c Q)^{-1} Q(2\mathcal{N}_2(r_c, w_2) + \mathcal{N}_3(r_c, r_c, r_c)), \quad (\text{G3})$$

and

$$\begin{aligned} w_4 = & -(QL_c Q)^{-1} Q \left(\mathcal{N}_2(w_2, w_2) + 2\mathcal{N}_2(r_c, w_3) \right. \\ & + 3\mathcal{N}_3(r_c, r_c, w_2) \\ & \left. + \mathcal{N}_4(r_c, r_c, r_c, r_c) \right). \end{aligned} \quad (\text{G4})$$

Substituting Eqs. (G2-G4) into the kernel equation yields the quintic normal form

$$\begin{aligned} \Psi_5(a; K) = & \lambda(K)a + \Lambda_3 a^3 + \Lambda_5 a^5 \\ & + O(a^7) + O(|K - K_c| a^3) = 0, \end{aligned} \quad (\text{G5})$$

where Λ_3 is still given by Eq. (D10), and

$$\begin{aligned} \Lambda_5 = & \left\langle r_c^\dagger, \mathcal{N}_5(r_c, r_c, r_c, r_c, r_c) \right. \\ & + 4\mathcal{N}_4(r_c, r_c, r_c, w_2) + 3\mathcal{N}_3(r_c, r_c, w_3) \\ & + 3\mathcal{N}_3(r_c, w_2, w_2) + 2\mathcal{N}_2(r_c, w_4) \\ & \left. + 2\mathcal{N}_2(w_2, w_3) \right\rangle. \end{aligned} \quad (\text{G6})$$

Eq. (G1) removes the quadratic and quartic terms from the reduced scalar equation. Eq. (G1) does not require w_2 or w_4 to vanish. Therefore the hysteresis analysis only needs Eq. (G5) together with the sign condition $\Lambda_5 > 0$. The explicit form of Λ_5 changes, but the later formulas do not.

Eq. (G6) also shows why a universal topological sign rule for Λ_5 is unavailable in the present framework. Unlike Λ_3 , the quintic coefficient depends on the slaved corrections w_2, w_3, w_4 and on the higher multilinear forms $\mathcal{N}_2, \mathcal{N}_5$. These objects are fixed by the analytic structure of the transport law. They are not fixed by the zero set of u_c alone. One may, for example, add a local quintic perturbation proportional to $\langle r_c^\dagger, r \rangle^5 r_c$. Such a perturbation changes Λ_5 . It does not change L_c , the topology of M , or the cubic sign statement in Appendix E. Therefore topology alone does not determine the sign of the general coefficient Λ_5 .

To obtain a shorter coefficient formula, we now add one more simplification. We assume that the nonlinear Taylor expansion along the critical branch contains no even terms up to quintic order:

$$\mathcal{N}(K_c, r) = \mathcal{N}_3(r, r, r) + \mathcal{N}_5(r, r, r, r, r) + O(\|r\|_X^7). \quad (\text{G7})$$

Under Eq. (G7), Eq. (D10) simplifies because $w_2 = 0$. Eqs. (G2-G4) then reduce to

$$\begin{aligned} w(a; K) = & a^3 w_3 + O(a^5) + O(|K - K_c| a), \\ w_3 = & -(QL_c Q)^{-1} Q \mathcal{N}_3(r_c, r_c, r_c). \end{aligned}$$

In that case, Eq. (G6) becomes

$$\Lambda_5 = \left\langle r_c^\dagger, \mathcal{N}_5(r_c, r_c, r_c, r_c, r_c) + 3\mathcal{N}_3(r_c, r_c, w_3) \right\rangle. \quad (\text{G8})$$

Eq. (G5) is the quintic extension of Eq. (D9). Eq. (G8) is a compact special case of the general coefficient formula in Eq. (G6). Eq. (G5) is the appropriate local normal form when the cubic term is not sufficient to describe the stable large-amplitude branch.

We now assume $\lambda'(K_c) > 0$ and $\Lambda_3 < 0 < \Lambda_5$. This is the subcritical situation selected by Appendix E when $\chi(M) \neq 0$ and the quintic term stabilizes the large-amplitude branch. For nontrivial equilibria, Eq. (G5) reduces to

$$\lambda(K) + \Lambda_3 x + \Lambda_5 x^2 = 0, \quad x := a^2 > 0. \quad (\text{G9})$$

Therefore the two nonzero branches are

$$x_\pm(K) = \frac{-\Lambda_3 \pm \sqrt{\Lambda_3^2 - 4\Lambda_5 \lambda(K)}}{2\Lambda_5}. \quad (\text{G10})$$

The reduced dynamics in Appendix F gives

$$\frac{d}{dt}a = -\Psi_5(a; K) + O(a^7) + O(|K - K_c| a^3). \quad (\text{G11})$$

Linearizing Eq. (G11) along a nonzero branch gives

$$\partial_a \Psi_5(a; K) = 2a^2 (\Lambda_3 + 2\Lambda_5 a^2) + O(a^6). \quad (\text{G12})$$

Hence the branch $x_+(K)$ is locally stable and the branch $x_-(K)$ is locally unstable at leading order.

The saddle-node point is determined by $\Lambda_3 + 2\Lambda_5 a^2 = 0$. Therefore

$$x_{\text{sn}} = -\frac{\Lambda_3}{2\Lambda_5}, \quad \lambda_{\text{sn}} = \frac{\Lambda_3^2}{4\Lambda_5}. \quad (\text{G13})$$

Using Eq. (D11), we obtain the leading spinodal location

$$K_{\text{sn}} - K_c \sim \frac{\Lambda_3^2}{4\lambda'(K_c)\Lambda_5}. \quad (\text{G14})$$

This quantity is the local hysteresis width:

$$\Delta K_{\text{hys}} := K_{\text{sn}} - K_c \sim \frac{\Lambda_3^2}{4\lambda'(K_c)\Lambda_5}. \quad (\text{G15})$$

At $K = K_c$, the stable large-amplitude branch satisfies

$$|a_{\text{jump}}| = \sqrt{-\frac{\Lambda_3}{\Lambda_5}}. \quad (\text{G16})$$

This is the jump amplitude when the incoherent branch loses stability at $K = K_c$. At the saddle-node point, the synchronized branch disappears with the smaller amplitude

$$|a_{\text{sn}}| = \sqrt{-\frac{\Lambda_3}{2\Lambda_5}}. \quad (\text{G17})$$

Therefore, under the additional condition $\Lambda_5 > 0$, the conditional sign statement for Λ_3 obtained in Appendix E extends to a local description of the discontinuous regime: the sign condition gives $\Lambda_3 < 0$, while the positive quintic

coefficient fixes the width of the hysteresis window and the size of the order parameter jump.

The same quintic normal form also refines the continuous regime. If $\Lambda_3 > 0$ and $\lambda(K) < 0$, the small supercritical branch of Eq. (G9) satisfies

$$x_{\text{cont}}(K) = -\frac{\lambda(K)}{\Lambda_3} - \frac{\Lambda_5}{\Lambda_3^3} \lambda(K)^2 + O(\lambda(K)^3). \quad (\text{G18})$$

Therefore Λ_5 gives the first higher-order correction to the continuous branch. It does not change the generic mean-field exponents derived in Appendix F. Those exponents are still controlled by the cubic term. The role of Λ_5 becomes leading only when $\Lambda_3 = 0$. If that codimension-two situation occurs and $\Lambda_5 > 0$, the transition is still continuous, but the scaling changes to

$$|a_*| \sim \left(-\frac{\lambda(K)}{\Lambda_5}\right)^{1/4}, \quad |a| \sim \left|\frac{h}{\Lambda_5}\right|^{1/5}. \quad (\text{G19})$$

This is the usual tricritical scaling. In the present theory, topology does not force this tricritical point. Such a point can only arise when the cubic coefficient is not already fixed to a definite nonzero sign.

Appendix H: Critical defect count and charge balance

In this appendix, we record the counting argument used in Subsec. IIID. In the main text, the field u_c is read as the critical texture, its isolated zeros are the defect cores, and the local index is the corresponding defect charge. Let $p_1, \dots, p_{N_{\text{def}}}$ be the isolated zeros of the critical tangent field u_c . The Poincaré-Hopf theorem [34] gives

$$\sum_{\ell=1}^{N_{\text{def}}} \text{ind}_{p_\ell}(u_c) = \chi(M). \quad (\text{H1})$$

Assume that each defect core is nondegenerate. Then each local charge is equal to +1 or -1. Let N_+ and N_- denote the numbers of positive and negative defect cores. Eq. (H1) then becomes

$$N_+ - N_- = \chi(M). \quad (\text{H2})$$

Therefore the total number of simple critical defects satisfies

$$N_{\text{def}} = N_+ + N_- \geq |\chi(M)|. \quad (\text{H3})$$

This is the generic counting statement used in the main text.

When $\chi(M) = 0$, Eq. (H2) gives $N_+ = N_-$. Simple critical defects must therefore be charge neutral as a set. Under smooth parameter variation, the total index remains fixed, so simple defects can only be created or annihilated through events with zero net topological charge. When $\chi(M) \neq 0$, Eq. (H2) shows that a nonzero background charge remains in the critical texture. That net charge cannot be removed by local charge-neutral defect events.

Appendix I: Conditional accessibility of local phase transition scenarios

In this appendix, we collect the logical consequences of Appendices E-G. The statements below concern only the local phase transition scenarios captured by the one-mode cubic and quintic reductions stated there. They do not claim a classification of the full global dynamics away from threshold. When symmetry produces a multidimensional critical space, these statements apply after selecting an individual symmetry-broken critical branch. The generic continuous branch in Appendix F requires

$$\Lambda_3 > 0. \quad (\text{I1})$$

The tricritical continuous branch in Appendix G requires

$$\Lambda_3 = 0, \quad \Lambda_5 > 0. \quad (\text{I2})$$

The discontinuous branch with local hysteresis requires

$$\Lambda_3 < 0, \quad \Lambda_5 > 0. \quad (\text{I3})$$

Suppose that the sign condition of Appendix E holds. If $\chi(M) \neq 0$, then Appendix E gives

$$\Lambda_3 < 0. \quad (\text{I4})$$

Eq. (I4) is incompatible with Eqs. (I1, I2). Therefore, within the conditional one-mode normal form, a non-zero Euler characteristic removes the generic continuous and tricritical continuous scenarios. Under the additional analytic condition $\Lambda_5 > 0$, Eq. (I3) remains as the local scenario selected near threshold.

If $\chi(M) = 0$, the sign of Λ_3 is not fixed by topology. In that case topology gives no sign selection for Λ_3 . The continuous, tricritical continuous, and discontinuous scenarios are then distinguished by Eq. (I1), Eq. (I2), and Eq. (I3), respectively. Their actual realization is decided by the analytic coefficients of the reduced equation. This is the sense in which the topology of M enters only conditionally in the local phase transition classification.

Appendix J: Geometric and topological indicators for representative manifolds

In this appendix, we collect the representative-manifold calculations used in Sec. IV. We first compute the averaged tangent-projection factor $\kappa(M)$, which gives the geometric coefficient multiplying the coupling-induced source term in the linearized kinetic equation. We then collect the Euler characteristics and the defect-counting consequences obtained from Appendix H.

1. Averaged tangent-projection factors

a. Hypersphere. Let $M = S^{D-1} \subset \mathbb{R}^D$ with the standard embedding. The tangent projector at $\sigma \in S^{D-1}$

is

$$\mathbf{P}_\sigma^\perp = \mathbf{I}_D - \sigma\sigma^\top. \quad (\text{J1})$$

Its average over the sphere is

$$\bar{\mathbf{P}}_{S^{D-1}} := \frac{1}{\text{Vol}(S^{D-1})} \int_{S^{D-1}} \mathbf{P}_\sigma^\perp d\mu(\sigma). \quad (\text{J2})$$

Rotational invariance implies $\bar{\mathbf{P}}_{S^{D-1}} = c\mathbf{I}_D$ for some scalar c . Taking the trace of Eq. (J2) gives

$$cD = \text{tr}(\bar{\mathbf{P}}_{S^{D-1}}) = D - 1, \quad (\text{J3})$$

so

$$\bar{\mathbf{P}}_{S^{D-1}} = \frac{D-1}{D} \mathbf{I}_D, \quad \kappa(S^{D-1}) = \frac{D-1}{D}. \quad (\text{J4})$$

b. Flat torus. Let $M = \mathbb{T}^d = (S^1)^d$ with the standard product embedding

$$F(\theta_1, \dots, \theta_d) = (\cos \theta_1, \sin \theta_1, \dots, \cos \theta_d, \sin \theta_d). \quad (\text{J5})$$

The tangent projector is block diagonal,

$$\mathbf{P}_{(\theta_1, \dots, \theta_d)}^\perp = \text{diag}(\mathbf{P}(\theta_1), \dots, \mathbf{P}(\theta_d)). \quad (\text{J6})$$

where

$$\mathbf{P}(\theta) = \begin{pmatrix} \sin^2 \theta & -\sin \theta \cos \theta \\ -\sin \theta \cos \theta & \cos^2 \theta \end{pmatrix}. \quad (\text{J7})$$

Averaging Eq. (J7) over one circle gives

$$\frac{1}{2\pi} \int_0^{2\pi} \mathbf{P}(\theta) d\theta = \frac{1}{2} \mathbf{I}_2. \quad (\text{J8})$$

Therefore

$$\bar{\mathbf{P}}_{\mathbb{T}^d} = \frac{1}{2} \mathbf{I}_{2d}, \quad \kappa(\mathbb{T}^d) = \frac{1}{2}. \quad (\text{J9})$$

c. Real Stiefel manifold. Let $M = \text{St}(p, n) \subset \mathbb{R}^{n \times p}$ with $2 \leq p < n$, equipped with the canonical Frobenius metric. For $X \in \text{St}(p, n)$ and $Z \in \mathbb{R}^{n \times p}$, the orthogonal projection onto $T_X \text{St}(p, n) = \{\Xi : X^\top \Xi + \Xi^\top X = 0\}$ is

$$\begin{aligned} \mathbf{P}_X^\perp(Z) &= Z - X \text{sym}(X^\top Z) \\ &= Z - \frac{1}{2} X (X^\top Z + Z^\top X). \end{aligned} \quad (\text{J10})$$

Let

$$\bar{\mathbf{P}}_{\text{St}(p, n)} := \frac{1}{\text{Vol}(\text{St}(p, n))} \int_{\text{St}(p, n)} \mathbf{P}_X^\perp d\mu(X). \quad (\text{J11})$$

The canonical left-right action of $\text{SO}(n) \times \text{SO}(p)$ on $\mathbb{R}^{n \times p}$ implies

$$\bar{\mathbf{P}}_{\text{St}(p, n)}(Q_1 Z R) = Q_1 \bar{\mathbf{P}}_{\text{St}(p, n)}(Z) R \quad (\text{J12})$$

for all $Q_1 \in \text{SO}(n)$ and $R \in \text{SO}(p)$. Therefore $\bar{\mathbf{P}}_{\text{St}(p, n)}$ commutes with the standard tensor-product action of $\text{SO}(n) \times \text{SO}(p)$ on $\mathbb{R}^{n \times p}$. By Schur's lemma, it is therefore a scalar multiple of the identity on $\mathbb{R}^{n \times p}$. Since each tangent projector has trace $np - \frac{p(p+1)}{2}$, we obtain

$$\bar{\mathbf{P}}_{\text{St}(p, n)} = \frac{2n-p-1}{2n} \mathbf{I}_{np}, \quad \kappa(\text{St}(p, n)) = \frac{2n-p-1}{2n}. \quad (\text{J13})$$

d. Rotation group. Let $M = \text{SO}(n)$ with the canonical embedding in $\mathbb{R}^{n \times n}$ equipped with the Frobenius inner product. For $R \in \text{SO}(n)$ and $Z \in \mathbb{R}^{n \times n}$, the orthogonal projection onto $T_R \text{SO}(n) = R\mathfrak{so}(n)$ is

$$\mathbf{P}_R^\perp(Z) = R \text{skew}(R^\top Z) = \frac{1}{2} (Z - RZ^\top R). \quad (\text{J14})$$

Let

$$\bar{\mathbf{P}}_{\text{SO}(n)} := \frac{1}{\text{Vol}(\text{SO}(n))} \int_{\text{SO}(n)} \mathbf{P}_R^\perp d\mu(R). \quad (\text{J15})$$

Bi-invariance of the Haar measure gives

$$\bar{\mathbf{P}}_{\text{SO}(n)}(Q_1 Z Q_2) = Q_1 \bar{\mathbf{P}}_{\text{SO}(n)}(Z) Q_2 \quad (\text{J16})$$

for all $Q_1, Q_2 \in \text{SO}(n)$. Therefore $\bar{\mathbf{P}}_{\text{SO}(n)}$ commutes with the left-right action of $\text{SO}(n) \times \text{SO}(n)$ on $\mathbb{R}^{n \times n}$. By the real form of Schur's lemma, $\bar{\mathbf{P}}_{\text{SO}(n)}$ is therefore a scalar multiple of the identity on $\mathbb{R}^{n \times n}$. Since each tangent projector has trace $n(n-1)/2$, we obtain

$$\bar{\mathbf{P}}_{\text{SO}(n)} = \frac{n-1}{2n} \mathbf{I}_{n^2}, \quad \kappa(\text{SO}(n)) = \frac{n-1}{2n}. \quad (\text{J17})$$

e. Unitary group. Let $M = \text{U}(d)$ with the canonical embedding in $\mathbb{C}^{d \times d} \cong \mathbb{R}^{2d^2}$, equipped with the real Frobenius inner product $\langle X, Y \rangle = \text{Re tr}(X^\dagger Y)$. For $U \in \text{U}(d)$ and $Z \in \mathbb{C}^{d \times d}$, the orthogonal projection onto $T_U \text{U}(d) = U\mathfrak{u}(d)$ is

$$\mathbf{P}_U^\perp(Z) = U \text{skew}(U^\dagger Z) = \frac{1}{2} (Z - UZ^\dagger U). \quad (\text{J18})$$

Let

$$\bar{\mathbf{P}}_{\text{U}(d)} := \frac{1}{\text{Vol}(\text{U}(d))} \int_{\text{U}(d)} \mathbf{P}_U^\perp d\mu(U). \quad (\text{J19})$$

For any phase $\lambda \in \text{U}(1) \subset \text{U}(d)$, left invariance of the Haar measure gives

$$\int_{\text{U}(d)} UZ^\dagger U d\mu(U) = \lambda^2 \int_{\text{U}(d)} UZ^\dagger U d\mu(U). \quad (\text{J20})$$

Choosing λ with $\lambda^2 \neq 1$ shows that the integral in Eq. (J20) vanishes. Therefore

$$\bar{\mathbf{P}}_{\text{U}(d)} = \frac{1}{2} \mathbf{I}_{2d^2}, \quad \kappa(\text{U}(d)) = \frac{1}{2}. \quad (\text{J21})$$

f. Special unitary group. For $M = \text{SU}(d)$, the tangent space consists of traceless skew-Hermitian directions. The canonical projection is therefore obtained from Eq. (J18) by removing the trace component of $U^\dagger Z$ before applying the left translation by U . In the canonical embedding into $\mathbb{C}^{d \times d}$, the averaged tangent projector then separates the trace direction from its orthogonal complement. For this reason, the linear response is described by a representation-dependent operator rather than by a single scalar coefficient on the full ambient space. The special case $\text{SU}(2) \cong S^3$ is already covered by the hypersphere case.

g. Complex Grassmannian. Let $M = \text{Gr}_k(\mathbb{C}^n)$ with $1 \leq k \leq n-1$. We represent a point of M by the rank- k Hermitian projector P onto the corresponding k -plane, and we use the centered projector embedding $F(P) = P - \frac{k}{n} \mathbf{I}_n$ into the real vector space of traceless Hermitian matrices. For a traceless Hermitian matrix Z , the orthogonal projection onto the tangent space of the conjugacy orbit through P is

$$P_P^\perp(Z) = PZ + ZP - 2PZP. \quad (\text{J22})$$

Let

$$\bar{P}_{\text{Gr}_k(\mathbb{C}^n)} := \frac{1}{\text{Vol}(\text{Gr}_k(\mathbb{C}^n))} \int_{\text{Gr}_k(\mathbb{C}^n)} P_P^\perp d\mu(P). \quad (\text{J23})$$

The $U(n)$ -invariance of the Haar measure implies that $\bar{P}_{\text{Gr}_k(\mathbb{C}^n)}$ commutes with conjugation on the traceless Hermitian subspace. Therefore it is a scalar multiple of the identity on that subspace. Since the real dimension of $\text{Gr}_k(\mathbb{C}^n)$ is $2k(n-k)$ and the ambient real dimension is $n^2 - 1$, we obtain

$$\begin{aligned} \bar{P}_{\text{Gr}_k(\mathbb{C}^n)} &= \frac{2k(n-k)}{n^2-1} \mathbf{I}_{n^2-1}, \\ \kappa(\text{Gr}_k(\mathbb{C}^n)) &= \frac{2k(n-k)}{n^2-1}. \end{aligned} \quad (\text{J24})$$

h. Complex projective space. Since $\mathbb{CP}^m = \text{Gr}_1(\mathbb{C}^{m+1})$, Eq. (J24) gives

$$\kappa(\mathbb{CP}^m) = \frac{2}{m+2}. \quad (\text{J25})$$

i. Sphere product. Let $M = (S^m)^q \subset (\mathbb{R}^{m+1})^q = \mathbb{R}^{q(m+1)}$ with the product embedding $F(x_1, \dots, x_q) = (x_1, \dots, x_q)$. The tangent projector at $(x_1, \dots, x_q)$ is

$$P_{(x_1, \dots, x_q)}^\perp = \text{diag}(\mathbf{I}_{m+1} - x_1 x_1^\top, \dots, \mathbf{I}_{m+1} - x_q x_q^\top). \quad (\text{J26})$$

Using the sphere computation from Eqs. (J1-J4) on each factor, we obtain

$$\bar{P}_{(S^m)^q} = \frac{m}{m+1} \mathbf{I}_{q(m+1)}, \quad (\text{J27})$$

and therefore

$$\kappa((S^m)^q) = \frac{m}{m+1}. \quad (\text{J28})$$

In particular,

$$\kappa(S^{2m} \times S^{2m}) = \frac{2m}{2m+1}. \quad (\text{J29})$$

2. Euler characteristics and defect consequences

We next collect the Euler characteristics and the corresponding defect-counting consequences used in Sec. IV. These statements combine the standard Euler-characteristic identities with the counting result of Appendix H [34, 35, 72].

a. Hypersphere. For $M = S^{D-1}$, one has [72]

$$\chi(S^{D-1}) = 1 + (-1)^{D-1}. \quad (\text{J30})$$

Hence

$$\chi(S^{D-1}) = \begin{cases} 2, & D \text{ odd}, \\ 0, & D \text{ even}. \end{cases} \quad (\text{J31})$$

When D is odd, Appendix H gives net defect charge 2 and $N_{\text{def}} \geq 2$ on a generic branch. When D is even, a smooth defect-free critical texture is topologically allowed, and any simple defects must be charge neutral as a set.

b. Flat torus. For $M = \mathbb{T}^d = (S^1)^d$, one has [72]

$$\chi(\mathbb{T}^d) = \chi(S^1)^d = 0. \quad (\text{J32})$$

Therefore a smooth defect-free critical texture is topologically allowed. If simple defects are present, Appendix H gives zero net charge.

c. Real Stiefel manifold. For $M = \text{St}(p, n)$ with $2 \leq p < n$, the projection onto the first vector gives a fiber bundle [22, 72]

$$\text{St}(p-1, n-1) \longrightarrow \text{St}(p, n) \longrightarrow S^{n-1}. \quad (\text{J33})$$

The multiplicativity of the Euler characteristic for compact fiber bundles [72] therefore gives

$$\chi(\text{St}(p, n)) = \prod_{j=0}^{p-1} (1 + (-1)^{n-1-j}). \quad (\text{J34})$$

For every $2 \leq p < n$, this product contains a zero factor. Hence

$$\chi(\text{St}(p, n)) = 0, \quad 2 \leq p < n. \quad (\text{J35})$$

Therefore a smooth defect-free critical texture is topologically allowed on every such real Stiefel manifold. If simple defects are present, Appendix H gives zero net charge.

d. Rotation and unitary groups. Every positive-dimensional compact connected Lie group admits a nowhere-vanishing left-invariant vector field. Therefore Poincaré-Hopf gives [34, 64]

$$\chi(\text{SO}(n)) = 0, \quad \chi(\text{U}(d)) = 0, \quad \chi(\text{SU}(d)) = 0. \quad (\text{J36})$$

Manifold family	Unconditional topological defect constraint	Locality condition for the one-mode reduction	Sign condition for the cubic coefficient
S^{D-1}	If D is odd, Poincaré-Hopf gives $Q_{\text{def}} = 2$ and $N_{\text{def}}^{\text{min}} = 2$ for simple zeros. If D is even, the net defect charge is zero.	The locality hypotheses of Appendix D must be verified for the chosen rotation-field ensemble and for the selected critical branch.	For odd D , the signed dominance condition in Eq. (E6) must be verified before inferring the sign of Λ_3 . For even D , topology gives no sign constraint on Λ_3 .
$S^{2m} \times S^{2m}$	Poincaré-Hopf gives $Q_{\text{def}} = 4$ and $N_{\text{def}}^{\text{min}} = 4$ for simple zeros.	The locality hypotheses must be verified for the product embedding and for the selected order parameter branch.	The Euler characteristic fixes the net defect charge, but it does not by itself imply Eq. (E6). The signed dominance condition must be checked separately.
$\text{Gr}_k(\mathbb{C}^n)$	Poincaré-Hopf gives $Q_{\text{def}} = \binom{n}{k}$ and $N_{\text{def}}^{\text{min}} = \binom{n}{k}$ for simple zeros.	The locality hypotheses must be verified for the Plücker embedding subspace and for the selected critical branch.	Schubert-cell topology determines $\chi(M)$, but the local weights γ_ℓ and the remainder $\mathcal{R}_U$ in Eq. (E6) must be controlled separately.
$\mathbb{CP}^m$	Poincaré-Hopf gives $Q_{\text{def}} = m + 1$ and $N_{\text{def}}^{\text{min}} = m + 1$ for simple zeros.	The locality hypotheses must be verified for the projective embedding subspace and for the selected critical branch.	The Euler characteristic fixes the net defect charge, but Eq. (E6) remains an additional signed dominance condition.
$\mathbb{T}^d$	The Euler characteristic is zero. A defect-free critical texture is topologically allowed, and any simple defects must have zero total index.	The locality condition is required if one performs a local normal-form calculation for a specified drift ensemble and selected critical branch.	There is no topological sign constraint on Λ_3 . The cubic coefficient is determined by the analytic normal-form calculation.
$\text{St}(p, n)$	The Euler characteristic is zero. A defect-free critical texture is topologically allowed, and any simple defects must have zero total index.	The locality condition is model dependent and must be checked after a specific matrix-valued model and a specific critical branch have been chosen.	There is no topological sign constraint on Λ_3 . The sign must be obtained from the normal-form coefficients of the specified model.
$\text{SO}(n)$	The Euler characteristic is zero. A defect-free critical texture is topologically allowed, and any simple defects must have zero total index.	The locality condition is model dependent and must be checked after a specific rotation-group model and a specific critical branch have been chosen.	There is no topological sign constraint on Λ_3 . The sign must be obtained from the normal-form coefficients of the specified model.
$\text{U}(d)$	The Euler characteristic is zero. A defect-free critical texture is topologically allowed, and any simple defects must have zero total index.	The locality condition is model dependent and must be checked after a specific unitary-group model and a specific critical branch have been chosen.	There is no topological sign constraint on Λ_3 . The sign must be obtained from the normal-form coefficients of the specified model.

TABLE VII. Topological defect constraint and additional assumptions for the representative manifolds. The defect constraint is unconditional and follows from the Poincaré-Hopf theorem together with the Euler-characteristic calculations collected in this appendix. The locality and sign columns state the extra hypotheses needed before the last column of Table VI can be read as a local phase transition conclusion.

Hence both the rotation-group and unitary-group cases allow a smooth defect-free critical texture. The same is true for $\text{SU}(d)$. If simple defects are present, Appendix H gives zero net charge in all three cases.

e. Sphere product. For $M = (S^m)^q$, one has [72]

$$\chi((S^m)^q) = (1 + (-1)^m)^q. \quad (\text{J37})$$

If m is odd, then $\chi((S^m)^q) = 0$. A smooth defect-free critical texture is then topologically allowed, and any simple defects must be charge neutral as a set. If m is even, then $\chi((S^m)^q) = 2^q$. Appendix H then gives

$$N_{\text{def}} \geq 2^q. \quad (\text{J38})$$

In particular, every manifold of the form $S^{2m} \times S^{2m}$ has $\chi = 4$ and therefore carries at least four simple defect cores on a generic branch.

f. Complex Grassmannian. The complex Grassmannian $\text{Gr}_k(\mathbb{C}^n)$ admits a Schubert-cell decomposition with one cell for each k -element subset of [35] $\{1, \dots, n\}$. Every Schubert cell has even real dimension. Therefore the Euler characteristic equals the number of Schubert cells:

$$\chi(\text{Gr}_k(\mathbb{C}^n)) = \binom{n}{k}. \quad (\text{J39})$$

Appendix H then gives

$$N_{\text{def}} \geq \binom{n}{k}. \quad (\text{J40})$$

Hence the critical texture on $\text{Gr}_k(\mathbb{C}^n)$ must carry net defect charge $\binom{n}{k}$ and, on a generic branch, at least $\binom{n}{k}$ simple defect cores.

g. Complex projective space. Since [35] $\mathbb{CP}^m = \text{Gr}_1(\mathbb{C}^{m+1})$, Eq. (J39) gives

$$\chi(\mathbb{CP}^m) = m + 1. \quad (\text{J41})$$

Appendix H then gives

$$N_{\text{def}} \geq m + 1. \quad (\text{J42})$$

Hence the critical texture on $\mathbb{CP}^m$ must carry net defect charge $m + 1$ and, on a generic branch, at least $m + 1$ simple defect cores.

Appendix K: Interpretation of the representative manifold families

In this appendix, we give a family-by-family reading of Table VI. The purpose is not to replace the linear stability condition or the nonlinear normal-form calculation. Instead, we separate the two indicators that enter the argument: the averaged geometric projection, which fixes the geometric strength of the coupling-induced source term, and the Euler characteristic, which fixes the net topological charge of the critical texture.

Table VII should be read together with the last column of Table VI. The entries with nonzero Euler characteristic have an unconditional defect content, but their discontinuous local branch is conditional on the locality assumptions of Appendix D, the signed dominance condition of Appendix E, and the stabilizing quintic condition $\Lambda_5 > 0$. The entries with zero Euler characteristic have no topological sign constraint on Λ_3 ; their local branch must be obtained from the normal-form coefficients of the specified model.

a. Hyperspheres and even-sphere products. For the standard hypersphere $S^{D-1} \subset \mathbb{R}^D$, Appendix J gives

$$\kappa(S^{D-1}) = \frac{D-1}{D}. \quad (\text{K1})$$

This coefficient is the averaged tangent-projection factor carried by the coupling term. The complete critical coupling is still obtained from the spectral self-consistency condition after the intrinsic rotation fields have been specified. The same appendix gives

$$\chi(S^{D-1}) = 1 + (-1)^{D-1}. \quad (\text{K2})$$

Thus odd ambient dimension D gives net defect charge 2, whereas even ambient dimension gives zero net defect charge. Under the local hypotheses used for the one-mode reduction, this reproduces the topological part of

the known parity distinction for hyperspherical synchronization.

The product $S^{2m} \times S^{2m}$ separates the geometric coefficient from the Euler characteristic more clearly. We obtain

$$\kappa(S^{2m} \times S^{2m}) = \frac{2m}{2m+1}, \quad \chi(S^{2m} \times S^{2m}) = 4. \quad (\text{K3})$$

The geometry therefore fixes the strength with which the coupling drives the linearized perturbation, while the topology fixes a nonzero net defect charge. Under the same local hypotheses, the local normal form does not contain a generic continuous branch, and the discontinuous branch requires the stabilizing quintic condition $\Lambda_5 > 0$.

b. Complex Grassmannians and complex projective spaces. For the complex Grassmannian $\text{Gr}_k(\mathbb{C}^n)$, the Plücker embedding gives

$$\kappa(\text{Gr}_k(\mathbb{C}^n)) = \frac{2k(n-k)}{n^2-1}. \quad (\text{K4})$$

The Schubert-cell decomposition gives

$$\chi(\text{Gr}_k(\mathbb{C}^n)) = \binom{n}{k}. \quad (\text{K5})$$

Hence the critical texture carries net defect charge $\binom{n}{k}$ and, on a generic branch with simple defects, at least $\binom{n}{k}$ defect cores. The complex projective space is the special case $\mathbb{CP}^m = \text{Gr}_1(\mathbb{C}^{m+1})$. Its geometric and topological indicators reduce to

$$\kappa(\mathbb{CP}^m) = \frac{2}{m+2}, \quad \chi(\mathbb{CP}^m) = m+1. \quad (\text{K6})$$

These two examples show that the same framework applies to homogeneous phase spaces that are not hyperspheres.

c. Zero-Euler toroidal and matrix manifolds. The remaining examples in Table VI have zero Euler characteristic. For the vector-phase torus $\mathbb{T}^d$ and the unitary group $\text{U}(d)$, the averaged tangent-projection coefficient is $1/2$. For the real Stiefel manifold and the rotation group, Appendix J gives

$$\kappa(\text{St}(p, n)) = \frac{2n-p-1}{2n}, \quad \kappa(\text{SO}(n)) = \frac{n-1}{2n}. \quad (\text{K7})$$

The Euler-characteristic calculation in Appendix J gives

$$\chi(\mathbb{T}^d) = \chi(\text{St}(p, n)) = \chi(\text{SO}(n)) = \chi(\text{U}(d)) = 0. \quad (\text{K8})$$

Therefore these manifolds do not impose a nonzero net defect charge on the critical texture. A smooth defect-free critical texture is topologically allowed, and if simple defects appear their total charge must vanish. In these zero Euler characteristic cases, the local transition branch is decided by the coefficients of the reduced normal form rather than by the Euler characteristic alone.

-
- [1] Y. Kuramoto, Self-entrainment of a population of coupled nonlinear oscillators, in *International Symposium on Mathematical Problems in Theoretical Physics*, Lecture Notes in Physics, Vol. 39, edited by H. Araki (Springer, Berlin, 1975) pp. 420–422.
 - [2] F. A. Rodrigues, T. Peron, P. Ji, and J. Kurths, The Kuramoto model in complex networks, *Phys. Rep.* **610**, 1 (2016).
 - [3] B. Pietras and A. Daffertshofer, Network dynamics of coupled oscillators and phase reduction techniques, *Phys. Rep.* **819**, 1 (2019).
 - [4] C. Bick, M. Goodfellow, C. R. Laing, and E. A. Martens, Understanding the dynamics of biological and neural oscillator networks through exact mean-field reductions: a review, *J. Math. Neurosci.* **10**, 9 (2020).
 - [5] H. Sakaguchi and Y. Kuramoto, A soluble active rotator model showing phase transitions via mutual entertainment, *Prog. Theor. Phys.* **76**, 576 (1986).
 - [6] M. Breakspear, S. Heitmann, and A. Daffertshofer, Generative models of cortical oscillations: Neurobiological implications of the Kuramoto model, *Front. Hum. Neurosci.* **4**, 190 (2010).
 - [7] Z. Lu, K. Klein-Cardena, S. Lee, T. M. Antonsen, M. Girvan, and E. Ott, Resynchronization of circadian oscillators and the east-west asymmetry of jet-lag, *Chaos* **26**, 094811 (2016).
 - [8] D. Witthaut, F. Hellmann, J. Kurths, S. Kettemann, H. Meyer-Ortmanns, and M. Timme, Collective nonlinear dynamics and self-organization in decentralized power grids, *Rev. Mod. Phys.* **94**, 015005 (2022).
 - [9] S. Chandra, M. Girvan, and E. Ott, Continuous versus discontinuous transitions in the D-dimensional generalized Kuramoto model: Odd D is different, *Phys. Rev. X* **9**, 011002 (2019).
 - [10] M. Lipton, R. Mirollo, and S. H. Strogatz, The Kuramoto model on a sphere: Explaining its low-dimensional dynamics with group theory and hyperbolic geometry, *Chaos* **31**, 093113 (2021).
 - [11] W. Zou, S. He, D. V. Senthilkumar, and J. Kurths, Solvable dynamics of coupled high-dimensional generalized limit-cycle oscillators, *Phys. Rev. Lett.* **130**, 107202 (2023).
 - [12] J. Markdahl, J. Thunberg, and J. Goncalves, High-dimensional Kuramoto models on Stiefel manifolds synchronize complex networks almost globally, *Automatica* **113**, 108736 (2019).
 - [13] R. Tron, B. Afsari, and R. Vidal, Intrinsic consensus on $SO(3)$ with almost-global convergence, in *2012 IEEE 51st IEEE Conference on Decision and Control (CDC)* (2012) pp. 2052–2058.
 - [14] F. Golse and S.-Y. Ha, A mean-field limit of the Lohe matrix model and emergent dynamics, *Arch. Ration. Mech. Anal.* **234**, 1445 (2019).
 - [15] G. L. Buzanello, A. E. D. Barioni, and M. A. M. de Aguiar, Matrix coupling and generalized frustration in Kuramoto oscillators, *Chaos* **32**, 093130 (2022).
 - [16] M. A. M. de Aguiar, Generalized frustration in the multi-dimensional Kuramoto model, *Phys. Rev. E* **107**, 044205 (2023).
 - [17] A. Arvanitoyeorgos, *An Introduction to Lie Groups and the Geometry of Homogeneous Spaces*, Student Mathematical Library, Vol. 22 (American Mathematical Society, 2003).
 - [18] S. Fiori, Synchronization of first-order autonomous oscillators on Riemannian manifolds, *Discrete Contin. Dyn. Syst. B* **24**, 1725 (2018).
 - [19] S. Fiori, A control-theoretic approach to the synchronization of second-order continuous-time dynamical systems on real connected Riemannian manifolds, *SIAM J. Control Optim.* **58**, 787 (2020).
 - [20] J. M. Montenbruck, M. Burger, and F. Allgower, Synchronization of diffusively coupled systems on compact Riemannian manifolds in the presence of drift, *Syst. Control Lett.* **76**, 19 (2015).
 - [21] S. Fiori, Non-delayed synchronization of non-autonomous dynamical systems on Riemannian manifolds and its applications, *Nonlinear Dyn.* **94**, 3077 (2018).
 - [22] A. Edelman, T. A. Arias, and S. T. Smith, The geometry of algorithms with orthogonality constraints, *SIAM J. Matrix Anal. Appl.* **20**, 303 (1998).
 - [23] S.-Y. Ha, D. Ko, and S.-Y. Ryoo, Emergent dynamics of a generalized Lohe model on some class of lie groups, *J. Stat. Phys.* **168**, 171 (2017).
 - [24] S.-Y. Ha, D. Ko, and S.-Y. Ryoo, On the relaxation dynamics of Lohe oscillators on some Riemannian manifolds, *J. Stat. Phys.* **172**, 1427 (2018).
 - [25] D. Kim and J. Kim, Low-dimensional reduction of the non-Abelian quantum synchronization models on the unitary group, *Kinet. Relat. Models* **17**, 436 (2023).
 - [26] D. Kim and J. Kim, Emergent behaviours of a non-abelian quantum synchronisation model over the unitary group, *Eur. J. Appl. Math.* **36**, 328 (2025).
 - [27] S.-Y. Ha and H. Park, From the Lohe tensor model to the Lohe hermitian sphere model and emergent dynamics, *SIAM J. Appl. Dyn. Syst.* **19**, 1312 (2020).
 - [28] S.-Y. Ha, M. Kang, and H. Park, Emergent dynamics of the Lohe hermitian sphere model with frustration, *J. Math. Phys.* **62**, 10.1063/5.0038769 (2021).
 - [29] S.-Y. Ha, M. Kang, and H. Park, Collective behaviors of the Lohe hermitian sphere model with inertia, *Commun. Pure Appl. Anal.* **20**, 2613 (2021).
 - [30] S.-Y. Ha and H. Park, On the gradient flow formulation of the Lohe matrix model with high-order polynomial couplings, *J. Stat. Phys.* **184**, 10.1007/s10955-021-02804-3 (2021).
 - [31] S.-Y. Ryoo, Asymptotic formation and orbital stability of phase-locked states in Kuramoto–Lohe type synchronization models on lie groups, *Commun. Math. Sci.* **23**, 1221 (2025).
 - [32] M. A. Lohe, Systems of matrix Riccati equations, linear fractional transformations, partial integrability and synchronization, *J. Math. Phys.* **60**, 072702 (2019).
 - [33] S.-Y. Ha, D. Kim, H. Park, and S.-Y. Ryoo, Constants of motion for the finite-dimensional Lohe type models with frustration and applications to emergent dynamics, *Physica D* **416**, 132781 (2020).
 - [34] J. W. Milnor, *Topology from the Differentiable Viewpoint*, revised ed. (Princeton University Press, 1997).
 - [35] J. Harris, *Algebraic Geometry: A First Course*, Graduate Texts in Mathematics, Vol. 133 (Springer, 1992).
 - [36] J. M. Lee, *Introduction to Riemannian Manifolds*, 2nd ed., Graduate Texts in Mathematics, Vol. 176 (Springer,

- 2018).
- [37] J. Nash, The imbedding problem for Riemannian manifolds, *Ann. Math.* **63**, 20 (1956).
 - [38] J. M. Lee, *Introduction to Smooth Manifolds*, 2nd ed., Graduate Texts in Mathematics, Vol. 218 (Springer, 2012).
 - [39] J. Markdahl, J. Thunberg, and J. Goncalves, Almost global consensus on the n -sphere, *IEEE Trans. Autom. Control* **63**, 1664 (2017).
 - [40] S. Boccaletti, J. A. Almendral, S. Guan, I. Leyva, Z. Liu, I. Sendina-Nadal, Z. Wang, and Y. Zou, Explosive transitions in complex networks' structure and dynamics: Percolation and synchronization, *Phys. Rep.* **660**, 1 (2016).
 - [41] P. S. Skardal and A. Arenas, Higher order interactions in complex networks of phase oscillators promote abrupt synchronization switching, *Commun. Phys.* **3**, 10.1038/s42005-020-00485-0 (2020).
 - [42] A.-S. Sznitman, Topics in propagation of chaos, in *Ecole d'Ete de Probabilites de Saint-Flour XIX—1989*, Lecture Notes in Mathematics, Vol. 1464 (Springer, 1991) pp. 165–251.
 - [43] M. Sarkar and N. Gupte, Phase synchronization in the two-dimensional Kuramoto model: Vortices and duality, *Phys. Rev. E* **103**, 032204 (2021).
 - [44] Y. Rouzairé and D. Levis, Dynamics of topological defects in the noisy Kuramoto model in two dimensions, *Front. Phys.* **10**, 10.3389/fphy.2022.976515 (2022).
 - [45] Y. Rouzairé and D. Levis, Defect superdiffusion and unbinding in a 2d XY model of self-driven rotors, *Phys. Rev. Lett.* **127**, 088004 (2021).
 - [46] Y. Tian, Data and code repository for geometry- and topology-controlled synchronization phase transition on manifolds, <https://github.com/doloMing/Geometry-and-topology-controlled-synchronization-phase-transition-on-manifolds> (2026), numerical data and experiment code.
 - [47] K. Binder, Finite size scaling analysis of Ising model block distribution functions, *Z. Phys. B* **43**, 119 (1981).
 - [48] V. Buendia, Mesoscopic theory for coupled stochastic oscillators, *Phys. Rev. Lett.* **134**, 197201 (2025).
 - [49] S. Petkoski and A. Stefanovska, Kuramoto model with time-varying parameters, *Phys. Rev. E* **86**, 046212 (2012).
 - [50] M. L. Barabash, S. Petkoski, and A. Stefanovska, Homogeneous delays in the Kuramoto model with time-variable parameters, *Phys. Rev. E* **90**, 052903 (2014).
 - [51] P. Kundu, P. Khanra, C. Hens, and P. Pal, Transition to synchrony in degree-frequency correlated Sakaguchi-Kuramoto model, *Phys. Rev. E* **96**, 052216 (2017).
 - [52] F. Battiston, G. Cencetti, I. Iacopini, V. Latora, M. Lucas, A. Patania, J.-G. Young, and G. Petri, Networks beyond pairwise interactions: Structure and dynamics, *Phys. Rep.* **874**, 1 (2020).
 - [53] A. Arnaudon, R. L. Peach, G. Petri, and P. Expert, Connecting Hodge and Sakaguchi-Kuramoto through a mathematical framework for coupled oscillators on simplicial complexes, *Commun. Phys.* **5**, 211 (2022).
 - [54] M. Nurisso, A. Arnaudon, M. Lucas, R. L. Peach, P. Expert, F. Vaccarino, and G. Petri, A unified framework for simplicial Kuramoto models, *Chaos* **34**, 053120 (2024).
 - [55] L. V. Gambuzza, F. Di Patti, L. Gallo, S. Lepri, M. Romance, R. Criado, M. Frasca, V. Latora, and S. Boccaletti, Stability of synchronization in simplicial complexes, *Nat. Commun.* **12**, 1255 (2021).
 - [56] L. Gallo, R. Muolo, L. V. Gambuzza, V. Latora, M. Frasca, and T. Carletti, Synchronization induced by directed higher-order interactions, *Commun. Phys.* **5**, 263 (2022).
 - [57] L. Neuhauser, A. Mellor, and R. Lambiotte, Multibody interactions and nonlinear consensus dynamics on networked systems, *Phys. Rev. E* **101**, 032310 (2020).
 - [58] A. P. Millan, H. Sun, L. Giambagli, R. Muolo, T. Carletti, J. J. Torres, F. Radicchi, J. Kurths, and G. Bianconi, Topology shapes dynamics of higher-order networks, *Nat. Phys.* **21**, 353 (2025).
 - [59] R. Berner, T. Gross, C. Kuehn, J. Kurths, and S. Yanchuk, Adaptive dynamical networks, *Phys. Rep.* **1031**, 1 (2023).
 - [60] I. Leyva, I. Sendina-Nadal, and S. Boccaletti, Explosive synchronization in mono and multilayer networks, *Discrete Contin. Dyn. Syst. B* **23**, 1931 (2018).
 - [61] X. Dai, X. Li, R. Gutierrez, H. Guo, D. Jia, M. Perc, P. Manshour, Z. Wang, and S. Boccaletti, Explosive synchronization in populations of cooperative and competitive oscillators, *Chaos Solitons Fractals* **132**, 109589 (2020).
 - [62] A. Bayani, S. Jafari, H. Azarnoush, F. Nazarimehr, S. Boccaletti, and M. Perc, Explosive synchronization dependence on initial conditions: The minimal Kuramoto model, *Chaos Solitons Fractals* **169**, 113243 (2023).
 - [63] J. Markdahl, D. Proverbio, L. Mi, and J. Goncalves, Almost global convergence to practical synchronization in the generalized Kuramoto model on networks over the n -sphere, *Commun. Phys.* **4**, 10.1038/s42005-021-00689-y (2021).
 - [64] B. C. Hall, *Lie Groups, Lie Algebras, and Representations: An Elementary Introduction*, 2nd ed., Graduate Texts in Mathematics, Vol. 222 (Springer, 2015).
 - [65] B. Coutinho, A. V. Goltsev, S. N. Dorogovtsev, and J. F. F. Mendes, Kuramoto model with frequency-degree correlations on complex networks, *Phys. Rev. E* **87**, 10.1103/physreve.87.032106 (2013).
 - [66] P. Ji, T. Peron, P. J. Menck, F. A. Rodrigues, and J. Kurths, Cluster explosive synchronization in complex networks, *Phys. Rev. Lett.* **110**, 218701 (2013).
 - [67] P. Ji, T. Peron, F. A. Rodrigues, and J. Kurths, Analysis of cluster explosive synchronization in complex networks, *Phys. Rev. E* **90**, 062810 (2014).
 - [68] Y. Maistrenko, B. Penkovsky, and M. G. Rosenblum, Solitary state at the edge of synchrony in ensembles with attractive and repulsive interactions, *Phys. Rev. E* **89**, 060901 (2014).
 - [69] H. Wu, L. Kang, Z. Liu, and M. Dhamala, Exact explosive synchronization transitions in Kuramoto oscillators with time-delayed coupling, *Sci. Rep.* **8**, 15521 (2018).
 - [70] A. P. Millan, J. J. Torres, and G. Bianconi, Complex network geometry and frustrated synchronization, *Sci. Rep.* **8**, 9910 (2018).
 - [71] M. Haragus and G. Iooss, *Local Bifurcations, Center Manifolds, and Normal Forms in Infinite-Dimensional Dynamical Systems*, Universitext (Springer, 2011).
 - [72] A. Hatcher, *Algebraic Topology* (Cambridge University Press, 2002).
 - [73] Y. Zhang, M. Lucas, and F. Battiston, Higher-order interactions shape collective dynamics differently in hypergraphs and simplicial complexes, *Nat. Commun.* **14**, 1605 (2023).

- [74] X. Liu, X. Li, and Y. Shi, Almost global synchronization of Kuramoto oscillators with symmetry breaking terms, *Sci. Rep.* **14**, 25304 (2024).
- [75] D. Levis, I. Pagonabarraga, and B. Liebchen, Activity induced synchronization: Mutual flocking and chiral self-sorting, *Phys. Rev. Research* **1**, 023026 (2019).
- [76] M. G. Crandall and P. H. Rabinowitz, Bifurcation from simple eigenvalues, *J. Funct. Anal.* **8**, 321 (1971).